

Software-as-a-Graph: Heterogeneous Graph Learning for Pre-Deployment Dependability Analysis of Asynchronous and Event-Driven Distributed Systems

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Abstract

Modern asynchronous publish-subscribe and microservice architectures pose significant deployment challenges. Without end-to-end visibility, such systems are susceptible to critical outages resulting from latent mismatches in middleware contracts, a phenomenon known as the Architecture-Code Gap. To facilitate pre-deployment reliability assessment without dependence on runtime telemetry, this work introduces Software-as-a-Graph (SaG), a static analysis framework that transforms Architecture-as-Code manifests into typed multigraphs with five principal entity types. The central claim is that two independent analytical pathways can assess reliability before deployment: a relation-specific Heterogeneous Graph Transformer with Quality-of-Service (QoS) edge encodings (HGT-QoS) to forecast cascade blast radii, and an interpretable, though unvalidated, ISO/IEC 25010/25019-based attribution layer that generates diagnostic remediation profiles.

Evaluation across twelve inductive scenarios and five open-source systems using simulation oracles demonstrates that relation typing and QoS encodings enhance rank correlation under distribution shift. However, relative to a training-free QoS-weighted centrality baseline, the learned model does not statistically improve ranking. Zero-shot transfer achieves an overall ρ of 0.767, which decreases to +0.265 for components that actively propagate failures and reverses on synchronous call trees. Forward inference requires 56 ms, while deterministic feature extraction results in cold static analysis being eleven times slower than direct simulation. These findings support the central claim that the model’s relational inductive bias and typed attention maps are advantageous under distribution shift.

Keywords: Heterogeneous graph neural networks, Distributed systems dependability, Publish-subscribe architecture, Cascading failures, Static system analysis, Explainable AI

1. Introduction

1.1. Motivation

Modern large-scale distributed software systems progressively employ asynchronous, event-driven, publish-subscribe (pub-sub) architectures. These architectures are widely used in domains such as autonomous driving (ROS 2 [1]), enterprise event streams (Apache Kafka [2]), cyber-physical systems (DDS [3]), IoT deployments (MQTT [4]), cloud-native microservices [5, 6], and distributed AI/LLM serving clusters. Pub-sub architectures decouple producers and consumers across space, time, and synchronization [7]. System components interact indirectly via message topics and brokers, removing the requirement for direct static references. Contemporary middleware specifications also allow deployment-time Quality-of-Service (QoS) policies, including reliability guarantees, durability, message priorities, and delivery deadlines, to manage system performance under peak load and network stress.

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Although this decoupling enables elastic scalability, it also creates a significant visibility barrier. Unlike synchronous architectures (e.g., RESTful HTTP, gRPC), which expose interactions as explicit caller–callee paths, publish–subscribe publishers and subscribers lack direct references. As a result, chain failures, head-of-line blocking, and backpressure may propagate along hidden logical paths involving brokers, shared topics, colocated hosts, and shared libraries [8, 9]. These failures primarily manifest through two mechanisms: sequential cascades, in which a slow subscriber saturates a broker queue and gradually throttles its publishers [10]; and simultaneous blast radii, where a shared library crash or host outage turns off all colocated services. Traditional architecture diagrams and static call graphs cannot capture these dissemination processes. Addressing such vulnerabilities is most effective before deployment, particularly during design and continuous integration, in accordance with dependable computing principles [11, 12]. At these stages, runtime telemetry, distributed tracing, and operational logs are unavailable. Therefore, architects and Site Reliability Engineers must identify systemically critical components, topics, and links, determine the underlying causes of their criticality, and select focused interventions such as broker replication, decoupling over-subscribed topics, or sandboxing shared libraries to reduce associated risks. In this context, systemic criticality refers to an entity’s propensity to function either as an architectural single point of failure, disrupting downstream connectivity upon failure, or as an error-propagation hub capable of initiating extensive multi-hop cascading outages within the asynchronous message mesh.

These considerations are also significant for computational sustainability. Architectural analysis based exclusively on manifests removes the necessity for provisioned clusters, active containers, or live fault-injection harnesses, thereby reducing deployment overhead and energy consumption. However, pre-deployment static analysis does not consistently surpass simulation in terms of speed: deterministic topological feature extraction requires 82.7s for a 520-component enterprise mesh and 239.3s for 2,000 components. In contrast, the in-process discrete-event cascade simulator completes in 0.14–7.2s (Sections 7.5.1 and 8.2). Although the learned GNN forward pass is minimal (56 ms), the overall static analysis pipeline does not confer a general computational advantage over in-process simulation. In continuous integration (CI/CD) workflows, realizing sustainability benefits requires caching deterministic graph metrics across commits and recomputing only the local subgraph affected by architectural pull requests.

1.2. Problem Statement: The Architecture–Code Gap and the Black-Box AI Challenge

Pre-deployment dependability and performance analysis is structured around two distinct, complementary tasks:

1. **Failure-Impact Forecasting (Predictive Pathway) — the primary task.** This task is the core claim: it forecasts dynamic cascading failure blast radii. It identifies components structurally critical to system-wide failure spread, using a data-driven, relation-specific model over learned topological representations. While closed-form topological metrics efficiently capture broad connectivity, their ability to resolve multi-hop, relation-dependent cascade spread across heterogeneous channels remains an empirical question, which we test directly against such a baseline (Section 7.1). We train and evaluate the predictive pathway against independent simulation ground truth as a ranking and critical-set identification model.
2. **Explainable Criticality Attribution (Explanation Layer) — tackling limitations of ranking alone.** While a ranked shortlist identifies where risk is concentrated, it does not indicate how to remediate it. Accordingly, the predictor is paired with an interpretable structural quality profile grounded in ISO/IEC 25010 [13] and ISO/IEC 25019 [14]. This layer diagnoses structural risk archetypes grounded in standardized software quality models—distinguishing, for example, an unreplicated single point of failure from an error-propagating cascade hub or a high-coupling maintainability bottleneck—to guide targeted architectural remediations. It functions strictly as an attribution model, not a ranking model.

This separation is architectural rather than solely presentational: both pathways operate on the same graph but do not share parameters, and neither is trained on the other’s output. The coupling term that could connect them is disabled by default and is reported only as an ablation (Section 4.2). This independence

enables Software-as-a-Graph (SaG) to identify components that are structurally central yet operationally low-impact, providing a nuanced diagnosis that neither pathway can achieve independently.

The distance between an architecture as designed and as realized is long-established: Perry and Wolf [15] named architectural erosion and drift three decades ago, and the architectural-technical-debt literature has tracked it since. What we label the **Architecture–Code Gap** is a specialization of that idea to asynchronous middleware, where the problem is not that an implementation diverged from its design but that the design’s failure semantics were never expressible in the artifacts a build pipeline can read. Existing software engineering approaches do not bridge it: *a distributed system can have pristine, bug-free source code within each service, yet remain fragile to critical global outages caused by hidden architectural single points of failure (SPOFs) or mismatched middleware Quality-of-Service (QoS) contracts*. This vulnerability is especially acute in asynchronous pub-sub architectures, where publishers and subscribers interact without direct static references, in sharp contrast to synchronous RPC call trees where exceptions bubble along explicit caller–callee edges. Classical architecture evaluation such as ATAM [16, 12], and the literature on architectural technical debt [17] and bad smells [18], identify architectural risks but rely on manual stakeholder elicitation rather than quantitative structural analysis. The automated paradigms each leave a different part of the gap unaddressed: static code analysis [19, 20, 21, 22] cannot see message queues or cross-host propagation; chaos engineering [23] needs a provisioned cluster and arrives after the architecture is fixed; and homogeneous centrality [24, 25, 26, 27] flattens the system into an untyped graph in which a topic, a library and a host are indistinguishable. Section 2 develops each in turn.

Furthermore, while machine learning has achieved notable success in software engineering, contemporary AI approaches to system dependability often function as uninterpretable black boxes. Deep neural models typically generate scalar risk scores or latent embeddings without providing transparent, actionable rationales for their predictions. In mission-critical software engineering, this lack of interpretability is insufficient; developers and Site Reliability Engineers need clear explanations of component vulnerabilities and compromised architectural mechanisms to refactor code or reconfigure infrastructure effectively.

1.3. The Software-as-a-Graph (SaG) Approach

To address both the Architecture–Code Gap and the black-box AI challenge, this work puts forward **Software-as-a-Graph (SaG)**, an AI-driven pre-deployment **Static System Analysis (SSA)** framework for asynchronous and event-driven distributed systems. SaG processes Architecture-as-Code expresses through a unified four-stage pipeline: (1) formulating the architecture as a typed, directed multigraph over Applications, Brokers, Topics, Execution Hosts, and Shared Libraries (Section 3.1); (2) projecting explicit physical connections into a QoS-weighted semantic **DEPENDS_ON** layer capturing sequential cascades and simultaneous blasts (Section 3.2); (3) training a Heterogeneous Graph Transformer (HGT) that forecasts multi-hop cascading blast radii and per-relationship criticality (Section 4); and (4) evaluating an explainable, standards-grounded Reliability–Maintainability (RM) attribution profile (Section 5) to pinpoint why components are fragile.

Crucially, SaG enforces a strict **input–label independence guarantee**: learned models and attribution baselines operate exclusively on the analytical graph G_{analysis} . At the same time, ground-truth failure impacts are generated by independent discrete-event simulators operating on the raw structural topology $G_{\text{structural}}$ (Section 4.4). Section 3 formalizes the complete architectural flow and pipeline interactions (Figure 1).

Rationale for Graph Learning vs. Direct Simulation. Although discrete-event simulation $I^*(v)$ defines ground-truth criticality and completes in 0.14–7.2 s, it is necessary to clarify the rationale for training a graph model. From a sustainable environmental management perspective, pre-deployment static manifest analysis eliminates the substantial carbon footprint associated with provisioning physical staging clusters, container fleets, and live chaos injection harnesses. However, as reported in Section 7.5.1, cold static feature extraction (82.7 s) on CPU does not outperform lightweight in-process simulation (7.2 s). In continuous integration (CI/CD) pipelines, static graph learning achieves practical speedup through deterministic topology caching: by caching base graph metrics across commits and extracting feature deltas only for pull-request-modified subgraphs, the sub-second neural forward pass (56 ms) provides instantaneous feedback on every code commit without re-running global simulation sweeps. Additionally, message passing generalizes across both labeled

and unlabeled entities, enabling the scoring of entity types (such as unsimulated shared libraries or physical hosts) and relationship-level criticalities (I_{edge} , Eq. (9)) that node-level simulation sweeps cannot evaluate without combinatorial edge-severing passes. Empirical data do not support two further rationales: cascade simulation exhibits negligible stochasticity on this corpus (median test–retest 0.982), and the claim that simulation requires runnable containers is incorrect for **FaultInjector**, which operates directly on raw manifests. Section 7.1 evaluates whether graph learning offers ranking advantages over closed-form baselines.

1.4. Research Questions

This empirical study considers five research questions:

RQ1 (Predictive Efficacy): *How accurately does heterogeneous graph learning predict cascading failure impact and identify the critical component set, compared with traditional, non-learning network indicators?*

RQ2 (Value of Architectural Typing): *Does modeling distinct entity and dependency types yield better failure predictions than homogeneous graph models on architectures the model has never seen — and does whatever advantage it confers depend on what other relational signal the model already has?*

RQ3 (QoS Encoding and Robustness): *Do middleware Quality-of-Service contracts carry signal a purely structural score discards, does that signal compose with or substitute for architectural typing, do the framework’s simulation oracles agree with one another, and are the reported orderings robust to the free parameters of the scorer and of the ground truth?*

RQ4 (Real-World Generalization): *How effectively does the framework transfer zero-shot to authentic, real-world distributed systems across autonomous driving (ROS 2), cloud-native microservices, smart home IoT, and industrial edge computing?*

RQ5 (Analysis Cost): *What does pre-deployment analysis cost at CI/CD time, which pipeline stage dominates that footprint, and how does it compare against the discrete-event simulation it is intended to displace?*

1.5. Key Contributions

This paper makes four principal contributions:

1. **Heterogeneous Graph Learning for Pre-Deployment Dependability, and Its Limits:** A relation-specific Heterogeneous Graph Transformer that forecasts cascading blast radii from Architecture-as-Code manifests, with a 16-D edge feature vector carrying 7 QoS dimensions and auxiliary multi-task heads for component and relationship criticality (Section 4). Ablated separately under inductive distribution shift across twelve architectures, the transition to relation-specific HGT demonstrates consistent OOD generalization ($\Delta\rho = +0.234$ over an untyped baseline, $p = 0.0005$). However, typing and continuous QoS encodings act as empirical substitutes rather than complements. Against an unparameterized QoS-weighted centrality baseline, learned ranking does not establish a statistically significant advantage ($+0.085$, $p = 0.151$). We report the substitution effect and the empirical boundary as primary findings (Sections 7.1–7.2).
2. **A Formal Typed Architecture Model:** A multigraph representation that derives logical dependencies from physical pub-sub linkages and distinguishes sequential cascade propagation from simultaneous multi-consumer library failures (Section 3).
3. **A Standards-Grounded Explanation Layer (a design contribution, not a validated one):** An interpretable Reliability–Maintainability model based on ISO/IEC 25010/25019 that distinguishes single-point-of-failure exposure from error-propagation reach (Section 5). Its evidential status is explicitly stated: it achieves modest ranking performance ($\rho = 0.205$, below unweighted centrality on every fold), degree centrality outperforms it on the pooled detection benchmark, and its elicited AHP weights are anti-predictive against a uniform prior. This model presents a design for standards-grounded attribution (Section 8.4).

4. Empirical Benchmark, Real-World Transfer, and Cost Profile: A study across seventeen system architectures totaling 2,812 components: twelve synthetic topologies (2,461 components across LOSO folds) and five open-source reference systems (351 components) under strict graph-view separation. The pipeline cost is characterized, demonstrating that the neural model contributes to only 0.02% of runtime. In contrast, deterministic feature analysis dominates (82.7 s versus 7.2 s for simulation), so we withdraw the conference version’s computational-efficiency claim (Sections 6–7).

Relationship to the authors’ prior work. A previous conference paper [28] introduced the preliminary multi-graph formulation and deterministic quality model on synthetic topologies. This JSS manuscript substantially extends that work through incorporating the complete predictive HGT pathway with 16-dimensional QoS edge encoding and multi-task heads (Section 4); inductive LOSO cross-validation (Section 7.2); zero-shot evaluation throughout five open-source systems (Section 7.4); empirical cost and sustainability characterization (Section 7.5); multi-oracle convergent validity and graph-view separation (Sections 4.3–4.4); and system-wide sensitivity analyses (Section 7.3). We retained formalisms from the conference paper only in restructured portions of Sections 3 and 5.

1.6. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the SaG multigraph model and dependency projections. Section 4 details the Heterogeneous Graph Transformer and simulation oracles. Section 5 presents the ISO/IEC-grounded explanation layer. Section 6 outlines the experimental protocol, and Section 7 reports empirical results for RQ1–RQ5. Section 8 discusses practical consequences, sustainability, threats to validity, and limitations. Section 9 concludes.

2. Related Work

This study’s core claim is that integrating four foundational research domains—(1) dependability, performance, and sustainability in distributed software systems; (2) static code and system analysis; (3) software quality analysis and multi-criteria evaluation; and (4) graph representation learning and explainable artificial intelligence (XAI)—enables prediction of systemic cascading vulnerabilities and performance degradation from Architecture-as-Code descriptors before runtime infrastructure provisioning.

2.1. Dependability, Performance, and Sustainability in Distributed Software Systems

The publish—subscribe (pub-sub) and asynchronous event-driven paradigms decouple communicating entities in space, time, and synchronization, enabling elastic scalability and high throughput [7]. Contemporary middleware standards, including ROS 2 [1], Apache Kafka [2], DDS [3], and MQTT [4], regulate these exchanges through fine-grained Quality-of-Service (QoS) policies that control message durability, transport reliability, priorities, and delivery deadlines. In cloud-native microservice meshes and distributed AI/LLM serving infrastructures, asynchronous message passing and queueing topologies constitute the primary communication substrate, directly influencing tail latencies, throughput bottlenecks, and hardware utilization.

Previous research on dependability and performance has primarily emphasized runtime mechanisms, such as dynamic consensus methods, broker clustering, adaptive backpressure throttling, autoscaling, and automated failover. Concurrently, chaos engineering and runtime verification [23] introduce faults or latency into staging or production clusters to observe system degradation and recovery. Although runtime fault injection provides operational validation that static methods cannot achieve, it requires a fully provisioned cluster, risks actual service disruption, and consumes significant computational resources. This situates it, along with model training, among development-time computations whose energy costs should be explicitly considered in green software engineering [29, 30, 31]. Consequently, such approaches are impractical during architectural design or lightweight commit-level continuous integration and deployment (CI/CD).

The present study addresses the pre-deployment phase by predicting systemic cascading vulnerabilities and performance degradation directly from Architecture-as-Code descriptors before runtime infrastructure provisioning. In the context of green software engineering, the analysis is conducted on manifests rather

than active deployments. The argument remains narrowly scoped to provisioning requirements rather than comparisons with alternative computational methods. Empirical results indicate that the static gate is more computationally demanding than the simulation oracle it was intended to replace (Sections 7.5.1 and 8.2). The avoidance of production restart storms is provided as a motivating rationale.

Architectural Reliability Prediction. Predicting dependability from architectural descriptions has a rich analytical lineage. Cheung’s absorbing-Markov-chain model [32] derives system reliability from component reliabilities and transfer-of-control graphs; Goseva-Popstojanova and Trivedi [33] systematize subsequent state-based, path-based, and additive families, surveyed by Immonen and Niemelä [34]. Model-driven frameworks, such as the Palladio Component Model [35] and layered queueing networks [36], predict performance and reliability from parameterized component specifications, while annotation-based approaches such as the AADL Error Model Annex [37] generate fault trees from declared error states. These methods answer broader questions than ours, but require pre-calibrated operational profiles and failure rates unavailable at commit time. SaG, by contrast, asks one targeted question from manifests alone: which components’ failures propagate furthest through the declared topology?

Data-Driven Failure Prediction and Root-Cause Analysis in Microservices. A substantial literature localizes faults in microservices from operational telemetry: Seer [38] and Sage [39] predict QoS violations from hardware counters and traces; MicroRCA [40] and TraceRCA [41] isolate root causes over service-dependency graphs; and DeepTraLog [42] and Eadro [43] apply graph neural networks to multimodal traces, logs, and metrics (surveyed across 98 papers by Zhang et al. [44]). In an industrial benchmark study, Zhou et al. [45] show that cascading outages in synchronous microservices stem from thread-pool exhaustion, RPC timeouts, and recursive retry storms propagating along call trees. In contrast, pub-sub failures propagate via broker queue saturation and message starvation. Crucially, all these approaches require a running cluster emitting runtime telemetry; SaG addresses the pre-deployment complement, operating on static manifests before code execution.

2.2. Static Code Analysis (SCA) vs. Static System Analysis (SSA)

Traditional **Static Code Analysis (SCA)** tools (e.g., SonarQube [19]) inspect source code Abstract Syntax Trees (ASTs) within individual services. They evaluate cyclomatic complexity [20], class cohesion, module coupling (e.g., Lack of Cohesion in Methods [LCOM], Coupling Between Objects [CBO]) [21, 22], and code duplication to flag internal code smells and defect-prone modules [46, 47, 48, 49]. However, SCA cannot observe runtime communication topology: it is blind to inter-service messaging channels, message broker queue saturation, and cross-host failure propagation.

Static recovery of system-level structure is an active area of complementary research. Bushong et al. [50] derive communication diagrams from static code analysis, and a recent review evaluates nine architecture recovery tools [51]. This body of literature aims to construct accurate representations of implemented systems to facilitate comprehension and drift detection. By contrast, SaG employs a declared topology as input to forecast cascade blast radii.

To bridge this “Architecture–Code Gap,” **Static System Analysis (SSA)** extends static analysis from single-service source code to the global system architecture. Using models of distributed applications, message topics, brokers, execution nodes, and shared libraries as a connected multigraph, SSA propagates code-level quality metrics across architectural dependencies. This allows engineering teams to detect structural anti-patterns [52, 53] and architectural technical debt [54] early during continuous integration (CI/CD) [55, 56], before defective topologies enter production.

2.3. Software Quality Models and Multi-Criteria Evaluation

Software product quality is standardized by the **ISO/IEC 25010:2023** product quality model [13] and the **ISO/IEC 25019:2023** Quality-in-Use model [14]. ISO/IEC 25010:2023 defines three closely intertwined characteristics: **Reliability** (comprising Faultlessness, Availability, Fault Tolerance, and Recoverability), **Maintainability** (Modularity, Reusability, Analyzability, Modifiability, and Testability), and **Efficiency of Performance** (Time Behavior, Resource Utilization, and Capacity). SaG operationalizes a strict subset

derivable from deployment topology: Availability and Fault Tolerance under Reliability, and Modularity, Modifiability, and Analyzability under Maintainability (Section 5.1). Faultlessness, Recoverability, Reusability, and Testability fall outside the scope of topological analysis.

Software engineering measurement explicitly distinguishes between *internal quality* (inbuilt structural attributes assessed on static artifacts at rest) and *external quality* (runtime dependability and behavioral characteristics noted during system execution) [57, 58]. In distributed architectures, architectural technical debt (such as over-centralized message topics or unreplicated brokers) degrades internal quality and precipitates severe external performance bottlenecks, queue congestion, and outages.

Aggregating multi-attribute structural metrics into an auditable quality score presents a classic Multi-Criteria Decision Making (MCDM) challenge. The Analytic Hierarchy Process (AHP) [59] provides a structured pairwise-comparison methodology with an explicit Consistency Ratio ($CR \leq 0.10$) to ensure coherence among elicited judgments. While this statistic detects inconsistency, it does not identify matrices completed from predetermined answers, a limitation we address for the weights in this study and quantify in Supplementary Section S4. The present work applies AHP to establish an audited, explainable Reliability–Maintainability (RM) quality baseline in conjunction with learned graph models.

2.4. Graph Representation Learning and Explainable AI

Network science offers established centrality indices for identifying critical nodes, such as degree, closeness, betweenness centrality [24, 26], articulation points, and PageRank [25, 27]. Foundational studies on network robustness [10], cascading overloads [8], and interdependent networks [9] model disruption propagation across interconnected topologies. While percolation models provide natural comparators, the training-free baselines in this study are centrality-based (Section 6.2). Evaluating targeted percolation fragmentation is identified as a direction for future baseline comparisons.

Standard network measures present two significant limitations in the context of software architectures. First, Dimensional Collapse arises when a single centrality value does not distinguish the underlying reasons for a component’s criticality, such as differentiating between an isolated single point of failure, an error-propagating cascade hub, or an over-shared library. Second, Semantic Collapse occurs when unweighted metrics treat all nodes and edges equivalently, conflating fundamentally distinct architectural entities, such as asynchronous message topics, shared libraries, and physical execution hosts.

To overcome the limitations of hand-engineered metrics, recent research applies machine learning to network vulnerability analysis (e.g., FINDER [60], DrBC [61], PowerGraph [62]). However, most models employ homogeneous message passing (GCN [63], GraphSAGE [64], GAT [65]) and indiscriminately average signals across connection types. Given the inherent heterogeneity of distributed software architectures, such models obscure entity boundaries and fail to generalize to out-of-distribution scenarios. Heterogeneous Graph Neural Networks (RGCN [66], HAN [67], HGT [68], MAGNN [69]) address this issue through relation-specific transformations. This study leverages the Heterogeneous Graph Transformer (HGT) [68] to preserve typed relational semantics when forecasting cascade blast radii. Graph learning has also been directly applied to microservice topologies; for instance, Khodabandeh et al. predict future service interactions using graph attention over temporally segmented call graphs. However, that approach forecasts edge existence based on observed interaction history. In contrast, the current method accepts a declared topology as input and predicts the blast radius resulting from node removal.

Explainable AI (XAI) vs. The Black-Box Barrier. A major challenge in applying modern artificial intelligence to software engineering is the black-box barrier: deep neural models generate risk scores or continuous embeddings without elucidating underlying structural causality. In production environments, these uninterpretable risk rankings hinder actionable decision-making, as developers and site reliability engineers (SREs) cannot determine whether to replicate hosts, configure circuit breakers, or refactor shared libraries.

Existing graph neural network (GNN) explanatory methods, such as GNNExplainer [70] and PGExplainer [71], identify influential subgraphs through edge masking or parameterized learning. While valuable, these approaches interpret models using internal hidden representations rather than standardized software engineering concepts. SaG addresses this limitation through a decoupled dual-pathway design. The predictive HGT pathway reveals typed mutual-attention distributions that indicate which architectural

relations propagated the cascade (Section 7.3.3 and Supplementary Section S8). Simultaneously, the deterministic explanation layer attributes fragility to standardized ISO/IEC quality sub-characteristics (Section 5), translating raw predictions into actionable, cost-effective remediations.

3. The Software-as-a-Graph (SaG) Architectural Model

Figure 1 presents the end-to-end architecture of the SaG framework. The shared front end processes Architecture-as-Code manifests, constructs a typed multigraph, projects implicit runtime interactions within a QoS-weighted logical dependency layer, and extracts typed node properties. These features are input to two independent pathways with no shared parameters: the predictive pathway (Section 4), which forecasts cascading failure blast radii and per-relationship criticality using a Heterogeneous Graph Transformer; and the explanation layer (Section 5), which decomposes fragility into Reliability and Maintainability quality profiles.

This section formalizes the Software-as-a-Graph multigraph representation (Section 3.1), the QoS-aware weighting and logical dependency derivation rules (Section 3.2), the dual graph views (Section 3.3), and the typed node feature encodings (Section 3.4).

3.1. Formal Multigraph Definition

A complex distributed software system is formally represented as a typed, weighted, directed multigraph:

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E) \quad (1)$$

where:

- V is the set of system entities, partitioned into five disjoint categories $\mathcal{T}_V = \{\text{app, broker, topic, host, lib}\}$ such that $V = \bigcup_{t \in \mathcal{T}_V} V_t$:

$$V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{host}} \cup V_{\text{lib}} \quad (2)$$

To prevent conflation of graph vertices with physical compute machines, V_{host} is designated as *Execution Hosts* (physical hosts or virtualized execution nodes).

- E is the set of directed edges connecting entities.
- $\tau_V : V \rightarrow \mathcal{T}_V$ and $\tau_E : E \rightarrow \mathcal{T}_E$ are typing functions assigning entity and relationship categories.
- $w_V : V \rightarrow (0, 1]$ and $w_E : E \rightarrow (0, 1]$ are weighting functions representing entity criticality and connection strength. For applications and shared libraries, $w_V(v)$ is initialized from static code metrics as $w_V(v) = 1 - \text{CQP}(v)$ (where CQP denotes the Code Quality Penalty, Section 3.4). It defaults to 1.0 when static code metrics are absent or for infrastructure entities ($w_V(\text{host}) = 1.0$).

Table 1 summarizes the five entity types and six structural edge types defined in the SaG model, together with their semantics and representative distributed-system implementations.

Application and Library entities also incorporate static code metrics generated by Static Code Analysis (SCA) tools (`cm_*` attributes: lines of code, cyclomatic complexity, coupling between objects, LCOM), thereby linking code-level fragility directly to topological analysis.

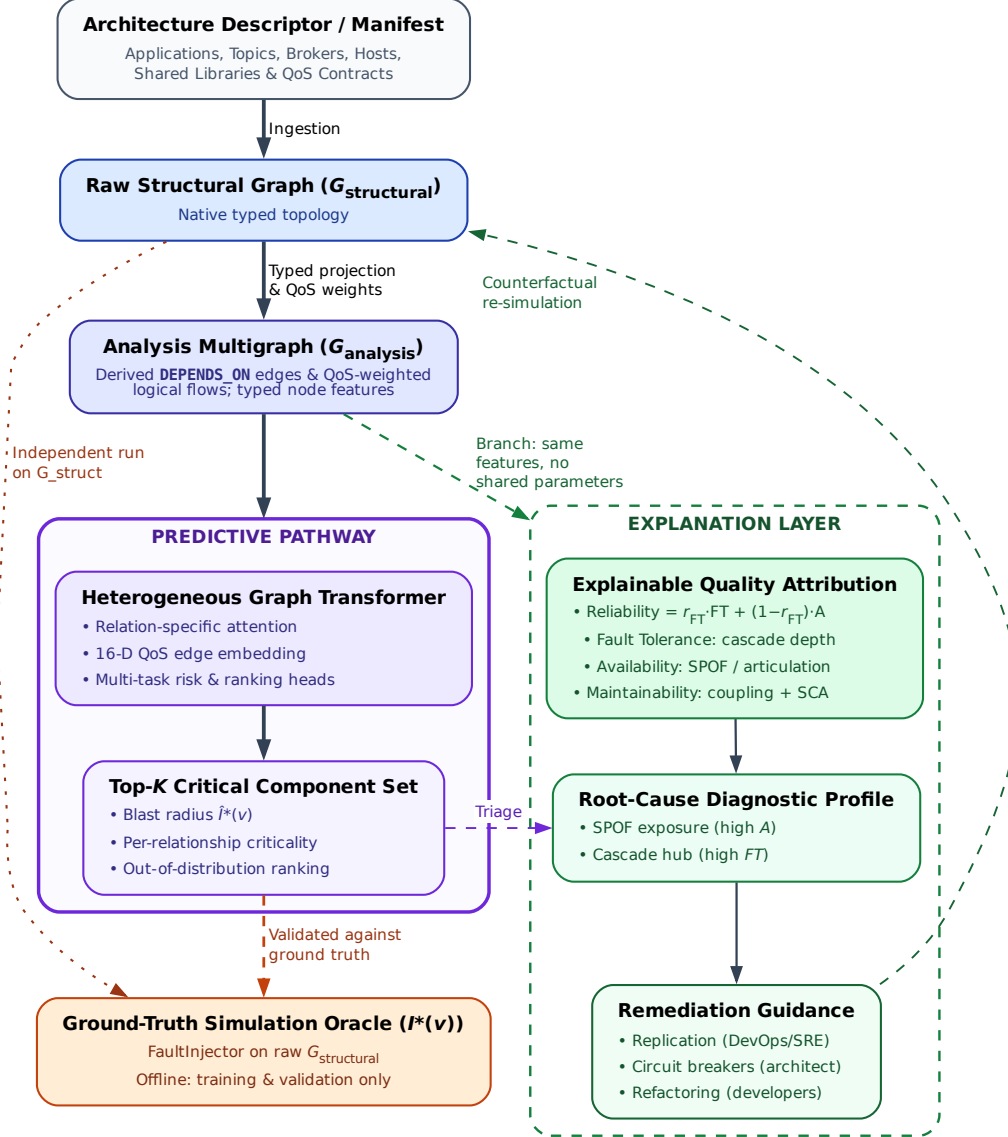


Figure 1: End-to-end architecture of the SaG framework. The central assertion is that the predictive pathway (Section 4) forms the primary sequence: manifest ingestion $\rightarrow$ typed multigraph $\rightarrow$ QoS-weighted **DEPENDS_ON** projection $\rightarrow$ typed node properties $\rightarrow$ heterogeneous graph learning $\rightarrow$ a ranked critical set with per-relationship criticality $\rightarrow$ the ground-truth simulation oracle (Section 4.3) that evaluates it. The oracle completes the predictive pathway’s training loop and operates solely on $G_{\text{structural}}$; it functions strictly offline and is excluded from inference (Section 4.4), as indicated by the dashed edge. The explanation layer (Section 5) diverges from this sequence: it re-enters from the analysis multigraph, generates a standards-based quality profile from the same typed features without sharing parameters with the predictor, and is accessed through triage rather than direct data flow.

Table 1: Entity and structural edge types in the SaG model.

Entity Type ($\mathcal{T}_V$)	Architectural Role	Concrete System Examples
Application (V_{app})	Autonomous process producing/consuming messages	ROS 2 node, Kafka microservice, MQTT client
Broker (V_{broker})	Message routing and queuing intermediary	RabbitMQ exchange, Mosquitto, EMQX broker
Topic (V_{topic})	Named logical communication channel	<code>/sensor/lidar</code> , <code>orders.payment.completed</code>
Execution Host (V_{host})	Physical host or virtualized execution environment	Bare-metal server, Kubernetes worker, Cloud VM
Library (V_{lib})	Shared software package or runtime dependency	<code>librdkafka</code> , OpenCV, Protobuf runtime
Structural Edge ($\mathcal{T}_E$)	Direction	Semantic Meaning
PUBLISHES_TO	App/Library $\rightarrow$ Topic	Component publishes messages to topic
SUBSCRIBES_TO	App/Library $\rightarrow$ Topic	Component consumes messages from topic
ROUTES	Broker $\rightarrow$ Topic	Broker manages and routes topic traffic
RUNS_ON	App/Broker $\rightarrow$ Host	Process is hosted on physical/virtual host
CONNECTS_TO	Host $\rightarrow$ Host	Physical network link between hosts
USES	App $\rightarrow$ Library	Application links to shared library dependency

3.2. QoS-Aware Weights and Logical Dependency Derivation

In distributed middleware, communication links differ in strength according to their Quality-of-Service (QoS) contracts. For example, a **RELIABLE** topic with **TRANSIENT_LOCAL** durability creates a stronger binding between communicating services than a **BEST_EFFORT** telemetry stream.

Each topic t carries an intrinsic criticality weight $w(t) \in (0, 1]$ combining its declared QoS semantics with two runtime-stress modulators: payload size and publication frequency:

$$w(t) = \alpha_{\text{top}} \cdot \text{QoS}(t) + \beta_{\text{top}} \cdot \text{SizeNorm}(t) + \gamma_{\text{top}} \cdot \text{FreqNorm}(t), \quad (\alpha_{\text{top}}, \beta_{\text{top}}, \gamma_{\text{top}}) = (0.75, 0.15, 0.10) \quad (3)$$

where the QoS term is an AHP-weighted aggregate of the declared contract:

$$\text{QoS}(t) = w_{\text{rel}} \cdot q_{\text{rel}} + w_{\text{dur}} \cdot q_{\text{dur}} + w_{\text{prio}} \cdot q_{\text{prio}}, \quad (w_{\text{rel}}, w_{\text{dur}}, w_{\text{prio}}) = (0.24, 0.62, 0.14) \quad (4)$$

Here, $q_{\text{rel}}, q_{\text{dur}}, q_{\text{prio}} \in [0, 1]$ represent normalized reliability, durability, and transport-priority scores mapped directly from declared manifest policies: $q_{\text{rel}} \in \{0.0, 1.0\}$ (best-effort vs. reliable), $q_{\text{dur}} \in \{0.0, 0.5, 1.0\}$ (volatile, transient-local, persistent), and $q_{\text{prio}} \in \{0.0, 0.5, 1.0\}$ (low, medium, high). Durability dominates because it governs whether data remains across restarts and network partitions. Reliability and transport priority both govern in-flight delivery quality, with reliability receiving higher weight because unconditional delivery guarantees precede message scheduling. The sub-weight vector is the geometric-mean priority vector of an independently stated Saaty pairwise-comparison matrix, printed with its consistency computation in Supplementary Section S4; $CR = 0.016$, small but non-zero.

The modulators $\text{SizeNorm}(t)$ and $\text{FreqNorm}(t)$ are logarithmically compressed and clamped to $[0, 1]$:

$$\text{SizeNorm}(t) = \min \left(1.0, \frac{\log_2(1 + B(t))}{20} \right), \quad \text{FreqNorm}(t) = \min \left(1.0, \frac{\log_{10}(1 + F(t))}{3} \right) \quad (5)$$

Here, $B(t)$ denotes the message payload size in bytes (with a 1 MiB design envelope, which represents the practical DDS sample ceiling before RTPS fragmentation becomes dominant), and $F(t)$ represents the nominal publication frequency in Hertz. The final weight $w(t)$ is clamped to $[0.01, 1]$ to ensure that best-effort edges remain visible during graph traversals. Each structural communication edge incident on t (**PUBLISHES_TO**, **SUBSCRIBES_TO**, **ROUTES**) inherits $w_E(e) = w(t)$ together with the topic's QoS vector.

The outer split $(\alpha_{\text{top}}, \beta_{\text{top}}, \gamma_{\text{top}})$ constitutes a declared convex combination. Comprehensive sensitivity analysis using Morris elementary effects (Supplementary Section S1) demonstrates that topic-weight variation alters downstream rank correlation by no more than 0.018 (with $\rho = 0.081$ maximum shift across the full simplex). This finding indicates that the split is a documented convention rather than a sensitive parameter.

3.2.1. Logical Dependency Projection (*DEPENDS_ON*)

Structural edges represent explicit deployment connections but do not capture implicit runtime dependencies. For instance, a subscriber depends on a publisher, yet no direct edge connects them in publish-subscribe architectures. To address this, a single unified semantic relation, *DEPENDS_ON*, is derived and directed from *dependent* to *dependency* ("if target fails, source is impacted"), following the six projection rules detailed in Table 2. The resulting weight $w \in (0, 1]$ quantifies the magnitude of operational coupling and indicates the conditional likelihood that a disruption in the dependency propagates to the dependent.

Table 2: The six *DEPENDS_ON* logical dependency projection rules.

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
1	app_to_app	Subscriber $\rightarrow$ Publisher (via shared Topic, incl. transitive USES)	$1 - \prod_{t \in T} (1 - w(t))$
2	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3	host_to_host	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	$\max_{u \in \text{hosted}(h_1), v \in \text{hosted}(h_2)} w_{\text{DEPENDS_ON}}(u \rightarrow v)$
4	host_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	$\max_{u \in \text{hosted}(h)} w_{\text{DEPENDS_ON}}(u \rightarrow b)$
5	app_to_lib	Application $\rightarrow$ Shared Library it USES	$H(w_V(\text{app}), w_V(\text{lib}))$
6	broker_to_broker	Broker $\leftrightarrow$ Broker (shared physical fault-domain colocation, symmetric)	$w_V(\text{host})$

Rules 1 and 2 aggregate the set of topics T connecting a component pair using a probabilistic union rather than a maximum [72, 73, 74]. This approach ensures that additional parallel failure vectors increase coupling monotonically while maintaining $w \in (0, 1]$. Rule 5 employs the harmonic mean $H(x, y) = 2xy/(x + y)$ [75] to combine the vertex weights of the consuming Application and the shared Library, thereby balancing caller and dependency criticality. Rules 3 and 4 extend application-level dependencies across host boundaries using the maximum coupling weight.

3.2.2. Sequential Cascades vs. Simultaneous Blasts

A central principle of the SaG model is the distinction between two degradation modes: (1) **Sequential Cascades (Rule 1)**, in which a failed publisher starves downstream subscribers sequentially through message queues and topic buffers; and (2) **Simultaneous Blasts (Rule 5)**, in which a crashed library or execution host causes all consuming applications and colocated brokers to fail instantaneously in a single shared-fate event. The preservation of entity types and relation-specific projection rules enables the SaG model to represent both mechanisms, whereas untyped homogeneous graphs collapse them into indistinguishable edges.

Rule 6 is the only symmetric projection rule: two brokers colocated on the same execution host share that host's physical failure domain (following the simultaneous-blast principle of Rule 5, with $w = w_V(\text{host})$). In production deployments, colocated brokers compete for CPU, memory, and I/O; a host crash halts all colocated instances simultaneously. Rule 6 does not model logical intra-cluster broker coupling (e.g., partition replication, quorum election, or shovel links, which do not require physical colocation). It applies in four of the eight detection benchmark scenarios (Supplementary Section S12) and contributes only 12 directed edges. Because simulation oracles operate strictly on $G_{\text{structural}}$ (Section 4.4), Rule 6 has zero influence on ground-truth failure labels.

3.3. Dual Graph Views and Architectural Layers

The SaG framework comprises two distinct representations: (1) **Structural Graph** ($G_{\text{structural}}$), which is the raw deployment graph containing physical and structural relations (PUBLISHES_TO, ROUTES, RUNS_ON, USES) and preserves the untransformed deployment topology; and (2) **Analysis Graph** (G_{analysis}), which is the projected graph containing derived *DEPENDS_ON* edges annotated with QoS weights and ingested SCA metrics. All GNN embeddings and analytical metrics are computed on the analysis graph (see running example in Supplementary Figure S14.1).

G_{analysis} is further organized into four analytical layers (Application, Middleware, Infrastructure, and Global System), which enables evaluation of criticality at subsystem levels in a manner consistent with hierarchical frameworks such as MIL-STD-498 [76].

3.4. Typed Node Feature Encoding

Within the SaG architecture, both the predictive pathway (Section 4) and the explanation layer (Section 5) utilize the same unified typed node properties from G_{analysis} . The predictive pathway projects these properties per entity type prior to heterogeneous message passing, while the explanation layer aggregates them into its quality profile. All five entity types share indices 0–17, which comprise a common block of topological metrics: in/out degree, betweenness, closeness, reverse PageRank, clustering coefficient, articulation score, and bridge load. These metrics are produced by the deterministic analysis stage, whose computational cost is characterized in Section 7.5. All topological metrics in this block are normalized to $[0, 1]$ within each graph: degrees are normalized by $|V| - 1$, betweenness and closeness follow standard network formulations, and reverse PageRank is normalized to unit sum. This normalization prevents raw graph size and component counts from dominating multi-layer perceptron projections during cross-scenario inductive transfer. Type-specific blocks extend the feature set to between 19 and 25 dimensions: source-code metrics and the Code Quality Penalty for Applications, two reverse-USES blast-radius drivers for Libraries, queue capacity for Brokers, publisher/subscriber counts and ordinal QoS criticality for Topics, and CPU and memory allocation for Execution Hosts (V_{host}). Supplementary Section S11 provides the index-by-index schema.

The shared block includes the graph composition, which is relevant for interpreting Section 7: betweenness, closeness, reverse PageRank, and articulation score are topological summaries computed prior to model evaluation. Consequently, a learned model is not the sole method for deriving a criticality score from structure, which ensures that the closed-form baselines serve as fair comparators rather than strawman alternatives.

4. Graph Learning for Failure-Impact Prediction

Cascading failure impact in distributed software systems exhibits non-linear, multi-hop, and relation-dependent characteristics. Outages propagate through architectural relations and dependencies that extend beyond immediate neighbors. The ability of a closed-form combination of standard centrality indices to capture these complex dynamics remains an open empirical question. Consequently, the primary predictive approach described in Section 1.2 employs a learned graph model, which is evaluated in Section 7.1 against a closed-form baseline. Notably, the learned model does not significantly outperform the baseline in out-of-distribution ranking.

This section presents the Heterogeneous Graph Transformer (HGT) architecture and its typed edge encodings (Section 4.1), the multi-task prediction heads and dimension-masked loss formulation (Section 4.2), the ground-truth simulation oracles (Section 4.3), and the input-label independence guarantee designed to prevent data leakage (Section 4.4).

4.1. Heterogeneous Graph Transformer Architecture

Because distributed systems comprise heterogeneous entity types (Applications, Libraries, Brokers, Topics, Execution Hosts) and diverse interaction semantics (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES, DEPENDS_ON), we employ a three-layer **Heterogeneous Graph Transformer (HGT)** architecture [68], implemented within PyTorch Geometric [77], with hidden dimension $D = 64$ and $H = 4$ attention heads. This architecture guarantees that typed relations, rather than simple adjacency, govern failure-impact forecasting.

4.1.1. Continuous-Categorical Edge Feature Encoding (16-D)

To capture continuous QoS constraints and channel semantics, SaG encodes each directed edge $e = (u, v)$ as a 16-dimensional continuous-categorical vector $e_{uv} \in \mathbb{R}^{16}$. Index 0 is the scalar coupling weight $w_E(e) \in (0, 1]$ of Section 3.2; index 1 is the normalized count of simple paths through e ; indices 2–8 one-hot encode the seven structural and derived relations; and indices 9–15 carry middleware QoS parameters on PUBLISHES_TO and SUBSCRIBES_TO edges, zeroed elsewhere. Six QoS dimensions are active in our corpus — reliability, durability, message priority, a heterogeneity flag raised when an edge’s QoS triple departs from its scenario’s modal profile, and the deadline pair (an active flag and $\log_{10}(1 + \text{deadline_ns}/10^6)$), populated on 463 of 615

topics 75%). The seventh, $\log_{10}(1 + \max_blocking_ms)$, is a schema provision for hard real-time DDS and ROS 2 profiles and is zero throughout.

An edge projection module maps e_{uv} into the hidden space: $e'_{uv} = W_{\text{edge}}e_{uv}$. Before relational attention computation, the current projection vector is incorporated directly into the target node representation: $\tilde{h}_v = h_v + e'_{uv}$.

4.1.2. Type-Specific Projection and Heterogeneous Message Passing

For each source node u and target node v connected by meta-relation $\tau(e) = (\tau(u), \phi(e), \tau(v))$, SaG follows the Heterogeneous Graph Transformer formulation of Hu et al. [68] implemented via PyTorch Geometric’s HGTConv [77]. Entity-specific projections $W_{\tau(v)}$ first map raw features $x_v \in \mathbb{R}^{19-25}$ into the shared D -dimensional hidden space: $h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v))$. Relational mutual attention across H heads incorporates type-parameterized Key ($K(u) = h_u^{(l-1)}W_K^{\tau(u)}$), Query ($Q(v) = \tilde{h}_v^{(l-1)}W_Q^{\tau(v)}$), and Value ($V(u) = h_u^{(l-1)}W_V^{\tau(u)}$) projections along with the edge representation $\tilde{h}_v = h_v + e'_{uv}$. Crucially, attention scores scale by a learned per-meta-relation prior $\mu_{\langle\tau(u), \phi(e), \tau(v)\rangle}$ (`p_rel` in PyG), which lets the model weight an entire relation triple up or down independently of individual node embeddings; this parameter directly captures the relational typing effect evaluated in Section 7.2. Message passing operates bidirectionally across both forward and transposed relation views (G_{analysis} and $G_{\text{analysis}}^\top$) to capture downstream starvation and upstream backpressure simultaneously, followed by residual aggregation, dropout ($p = 0.10$), and layer normalization across layers $l \in \{1, \dots, L\}$. Supplementary Section S1.1 provides the complete formal equation set.

Training Protocol and Optimization Hyperparameters. Models are optimized end-to-end using AdamW with initial learning rate $\eta = 3 \times 10^{-4}$, weight decay 10^{-4} , and dropout probability $p = 0.10$ applied post-attention. Learning rates follow a cosine decay schedule with warm restarts (CosineAnnealingWarmRestarts, $T_0 = 75$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 3 \times 10^{-6}$). Training runs for up to 300 epochs, with early stopping governed by a patience of 30 epochs on validation loss over labeled nodes. We process inductive subgraphs per scenario using full-graph inductive packing without mini-batch subsampling, with validation masks isolating held-out nodes to prevent information leakage across data splits. We evaluate five independent random seeds $\{42, 123, 456, 789, 2024\}$ across all runs, redrawing both partition masks and initializations. *Selection protocol:* the architectural hyperparameters ($D = 64$, $H = 4$, dropout, learning rate, and schedule) follow values conventional for HGT [68]. The loss coefficients in Equation (6) have no precedent—the objective is bespoke to this task—and we set them by judgment and left them untuned; we state this rather than appeal to a convention that does not exist for a five-term multi-task loss. Neither group was tuned against the in-distribution test split or the LOSO folds; we performed no search over them. The real-world evaluation in Section 7.4.1 is a separate case and is documented separately: it runs at a different depth and epoch budget from every other learned result in this paper, and Section 7.4.1 states the configuration and how we arrived at it. This avoids selection leakage, at the cost of leaving open whether either family is reported near its own optimum — a comparison between untuned configurations, which we state rather than treat as a like-for-like optimum comparison.

4.2. Multi-Task Prediction Heads and Dimension Masking

From the final node embeddings $h_v^{(L)}$, SaG utilizes specialized multi-task prediction heads:

- **Reliability Head:** $\hat{R}(v) = \sigma(\text{MLP}_R(h_v)) \in [0, 1]$
- **Maintainability Head:** $\hat{M}(v) = \sigma(\text{MLP}_M(h_v)) \in [0, 1]$
- **Composite Failure Impact Head:** $\hat{I}^*(v) = \sigma(\text{MLP}_C(h_v \parallel \hat{R}(v) \parallel \hat{M}(v))) \in [0, 1]$
- **Relationship Criticality Head:** $\hat{Q}(u, v) = \sigma(\text{TypedEdgeEncoder}_{\phi(e)}(h_u, h_v, e_{uv})) \in [0, 1]$

4.2.1. Dimension-Masked Loss Formulation

The combined optimization objective integrates regression accuracy, multi-task dimension learning, ranking fidelity, pairwise ordering, and edge prediction:

$$\mathcal{L} = \mathcal{L}_{\text{composite}} + 0.5 \cdot \mathcal{L}_{\text{dimension}} + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}} + 0.3 \cdot \mathcal{L}_{\text{edge}} + \lambda_{\text{RM}} \cdot \mathcal{L}_{\text{consistency}} \quad (6)$$

where $I^*(v)$ is the simulated cascade impact defined by the primary oracle (Section 4.3), $\mathcal{L}_{\text{composite}} = \text{MSE}(\hat{I}^*(v), I^*(v))$, $\mathcal{L}_{\text{rank}}$ is the ListMLE listwise ranking loss [78] parameterized by temperature τ :

$$\mathcal{L}_{\text{rank}} = -\frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{s}_{\pi_i}}{\tau} - \log \sum_{j=i}^N \exp \left(\frac{\hat{s}_{\pi_j}}{\tau} \right) \right) \quad (7)$$

where $\pi = (\pi_1, \dots, \pi_N)$ denotes the permutation of nodes sorted in descending order of ground-truth impact $I^*(v)$, and $\hat{s}_v = \hat{I}^*(v)$. At the baseline default $\tau = 1.0$, the formulation reduces to standard ListMLE; the temperature parameter $\tau < 1.0$ is a configurable hyperparameter that sharpens probability distributions over narrow prediction margins. Pairwise ordering fidelity is guided by margin-ranking loss $\mathcal{L}_{\text{pairwise}} = \frac{1}{|P|} \sum_{(u,v) \in P} \max(0, \gamma - (\hat{s}_u - \hat{s}_v))$ with margin $\gamma = 0.05$ over pairs $P = \{(u, v) \mid I^*(u) - I^*(v) > \gamma\}$, and $\mathcal{L}_{\text{consistency}} = \text{MSE}([\hat{R}(v), \hat{M}(v)]_{v \in \text{unlabeled}}, [R_{\text{RM}}(v), M_{\text{RM}}(v)]_{v \in \text{unlabeled}})$ regresses predicted heads toward the diagnostic pathway’s baseline on unlabeled nodes, where $R_{\text{RM}}(v)$ and $M_{\text{RM}}(v)$ denote the deterministic Reliability and Maintainability scores from the explanation layer (Section 5). Headline results use $\lambda_{\text{RM}} = 0$, guaranteeing that the predictive and elucidative pathways remain strictly independent.

The coefficients in Eq. (6) (0.5 dimension, 0.3 listwise rank, 0.1 pairwise margin, 0.3 edge) were chosen to focus on the primary composite regression while regularizing relative node rankings and edge classifications. Empirical testing sweeps confirmed stable convergence across all random seeds, with gradient norms remaining well-conditioned, thus preventing any individual objective from controlling the gradient.

Dimension Masking and Head Roles: Because dynamic cascade simulation ($I^*(v)$ via **FaultInjector**) observes runtime failure reachability rather than source-code maintainability, maintainability ground truth is unobserved during dynamic simulation. A separate change-propagation oracle $I_M(v)$ evaluates static structural change ripple at the Validate stage, but is never used as a training label to avoid circular supervision. We introduce a boolean dimension mask $m = [m_R, m_M] = [1, 0]$:

$$\mathcal{L}_{\text{dimension}} = \frac{1}{\sum_d m_d} \sum_{d \in \{R, M\}} m_d \cdot \text{MSE}(\hat{d}(v), d^*(v)) \quad (8)$$

This mask ensures the unobserved maintainability head is not artificially penalized or driven toward zero during backpropagation.

Auxiliary Nature of the Reliability Head: The surviving term deserves to be stated plainly, because it operates as an auxiliary feature pathway rather than multi-dimensional supervision. **FaultInjector** emits a single continuous scalar per component, and the label extractor assigns that same scalar to both the composite and reliability targets: $R^*(v) = I^*(v)$ identically. $\mathcal{L}_{\text{dimension}}$ under $m = [1, 0]$ therefore regresses $\hat{R}$ toward the exact same target $\mathcal{L}_{\text{composite}}$ regresses $\hat{I}^*$ toward. The two terms are not redundant — they train separate heads, and $\hat{R}$ re-enters the composite head as an input ($\hat{I}^* = \sigma(\text{MLP}_C(h_v \parallel \hat{R} \parallel \hat{M}))$), functioning as a feature-enrichment pathway rather than independent multi-task supervision. This oracle decomposes no second dimension of ground-truth; a distinct reliability score would require an independent oracle separating fault-tolerance from availability, which $I^*(v)$ does not do. We report the objective as implemented rather than claiming multi-dimensional supervisory ground truth.

4.2.2. Domain-Reweighted Criticality

ISO/IEC 25019’s Context of Use implies that the weight placed on reliability against maintainability is a deployment choice rather than a constant. The framework exposes a reweighting $\hat{Q}_{\text{domain}}(v) = q_R \hat{R}(v) + q_M M_{\text{static}}(v)$ to express it. Because maintainability is unobserved undergoing dynamic simulation ($m = [1, 0]$),

no headline result uses it: every reported figure is $\hat{I}^*(v)$ directly. Supplementary Section S4 reports its sensitivity against the static RM baseline.

4.3. Ground-Truth Simulation Oracles

To evaluate predictive accuracy before deployment without relying on production runtime telemetry, SaG executes discrete-event failure simulations over the raw structural multigraph $G_{\text{structural}}$. We create a formal taxonomy of four component-level oracles and one relationship-level oracle:

- **Cascade Reachability Oracle ($I^*(v)$)**, via **FaultInjector**: crashes component v , propagates outages across dependent topics, brokers, and links by breadth-first traversal, and returns the mean fractional feed loss over the subscriber population, each subscriber contributing the unweighted mean loss of the topics it subscribes to. A topic’s feed loss is the fraction of its publishers that have failed (for a topic with no publisher, the fraction of its failed routers), scaled by a QoS ladder ($\times 1.2$ RELIABLE, $\times 1.15$ high/urgent priority, $\times 1.05$ medium) and clamped to $[0, 1]$. The denominator is the full subscriber set of the intact graph, so subscribers that themselves fail are retained in the average rather than excluded. The implementation admits a per-publisher rate weighting, but rates are declared per topic throughout our corpus, so it reduces exactly to this publisher fraction. This is the **primary continuous target label** throughout.

How much QoS is in this label. The ladder reads reliability and transport priority only; durability does not enter I^* at all, despite carrying the largest of the three QoS sub-weights in the framework’s own elicited vector (0.62, against 0.24 for reliability as well as 0.14 for priority; Section 3.2). That omission does not limit the label’s QoS content. Re-running the labeler with QoS scaling disabled entirely leaves the Application ordering very nearly intact — mean Spearman $\rho = 0.965$ against the ladder across the twelve folds (range 0.891–0.999) — and substituting a durability-aware $w(t)$ scaling moves it less still ($\rho = 0.977$). Neither parameterization materially reorders the target. The top- K critical set is the more sensitive construct: ladder and topology-only labels agree at mean Jaccard 0.678, so QoS does change *which* components are named critical without changing their order. I^* should therefore be read as a near-topological target that carries a QoS term at its threshold boundaries rather than through its ranking, which bounds what any QoS-encoding result can be crediting (Section 7.3.1).

- **Multi-Metric Composite Oracle ($I_{\text{comp}}(v)$)**, via **FailureSimulator**: a severity-weighted mixture of reachability loss, fragmentation, throughput loss and flow disruption, with AHP-derived coefficients (0.35, 0.25, 0.25, 0.15). Those coefficients come from a rank-one comparison matrix, so they record their origin without independently justifying them. They are not swept in our sensitivity analysis — a gap worth naming because I_{comp} supplies the labels for the explanation layer’s real-world evaluation (Supplementary Sections S4 and S7). It is reserved for Validate-stage gates and prescriptive verification, never for forecasting ranking.
- **Dynamic Queue-Flow Oracle ($I_{\text{dyn}}(v)$)**, via **MessageFlowSimulator** on SimPy [79]: simulates emission rates, stochastic latencies, broker buffer saturation, and queue drops under fault injection, extracting the drop in delivered message rate to surviving consumers. It serves only as an independent convergent-validity probe (Supplementary Section S9).
- **Change-Propagation Oracle ($I_M(v)$)**, via **ChangePropagationSimulator**: a deterministic reverse-dependency traversal over the transpose of the six-rule DEPENDS_ON projection, blending change reach, weighted change impact, and normalized depth. It is a structural maintainability reference and is never used as a training label, which would make the supervision circular.
- **Relationship (Edge) Removal Oracle ($I_{\text{edge}}(u, v)$)**: the systemic impact of severing one dependency while both endpoints stay operational. Writing $\bar{I}_{\text{comp}}(G)$ for the mean composite impact over G :

$$I_{\text{edge}}(u, v) = \bar{I}_{\text{comp}}(G \setminus \{(u, v)\}) - \bar{I}_{\text{comp}}(G) \quad (9)$$

While Eq. (9) is formulated using $\bar{I}_{\text{comp}}$ in the multi-metric quality suite, I_{edge} can equivalently be defined with respect to the primary reachability oracle $I^*(v)$, measuring the change in mean subscriber feed loss when dependency (u, v) is severed.

Topic Criticality Label Masking. `FailureSimulator` can blend a declared `Topic.criticality` into its severity term; this is disabled because that field is a GNN input feature (`topic_qos_criticality_ord`) and consuming it would score the predictor against a transformation of its own input.

Primary Oracle Declaration and Role Assignment. Because the three reliability-facing oracles (I^* , I_{comp} , I_{dyn}) measure separate operational constructs, we designate $I^*(v)$ (**FaultInjector**) as the **primary continuous target oracle** for all predictive ranking results (Tables 4–5, RQ1–RQ3). We select $I^*(v)$ over $I_{\text{dyn}}(v)$ for two methodological reasons: first, deterministic cascade reachability isolates structural dependency propagation with zero seed-to-seed variance, providing the reproducible ground truth required for deterministic CI/CD regression gating; second, discrete-event queue simulation (I_{dyn}) introduces stochastic message latencies, bursty arrival distributions, and synthetic buffer limits that introduce queuing noise and workload assumptions, obscuring intrinsic architectural topology. $I_{\text{comp}}(v)$ is reserved for Validate-stage quality gates and prescriptive remediation verification, $I_{\text{dyn}}(v)$ serves as an independent convergent-validity probe (Section 7.3.2 and Supplementary Section S9), and $I_M(v)$ serves as a structural maintainability reference.

Cross-Oracle Convergent Validity. As detailed in Section 7.3.2 and Supplementary Section S9, the three reliability oracles show substantial but sub-ceiling agreement on Applications ($\rho = 0.620$ for (I_{dyn}, I^*) against a 0.811–1.000 label noise floor), confirming distinct constructs. Consequently, results established against one oracle are never transferred to another; every evaluation explicitly references its underlying simulation oracle.

4.4. Input-Label Independence Guarantee

To eliminate data leakage, SaG applies strict architectural separation: Feature Space is constructed exclusively from G_{analysis} using static structural topology, static code metrics, and declared QoS contracts, whereas Label Space is evaluated exclusively on raw $G_{\text{structural}}$ through independent simulation oracles (**FaultInjector**, **FailureSimulator**, **MessageFlowSimulator**). No simulation outputs, failure trace histories, or dynamic execution telemetry are ever exposed as input attributes to the GNN or the explanation layer.

What this guarantee does and does not establish. The separation rules out circular feature construction: no predictor can read a transformation of the quantity it is scored against. It does not establish probabilistic independence between features and labels, and we do not claim it does. G_{analysis} is a deterministic projection of $G_{\text{structural}}$ (Section 3.2). Hence, the labels are — up to the simulator seed — a deterministic function of the same topology from which the features are computed. Two consequences follow, and both bound the results of Section 7.

First, $I^*(v)$ is itself a topological functional: a breadth-first reachability computation over $G_{\text{structural}}$ scaled by a QoS ladder. The predictive task is therefore to recover a closed-form function of a graph from features of that same graph. Read that way, it is unsurprising that an unparameterized centrality score is competitive with a trained model (Section 7.1); the parity we report there is the expected outcome of the setup rather than a surprising failure of graph learning, and we flag it here so the reader does not have to infer it.

Second, and more restrictively, no result in this paper is validated against an observed failure. Every label — on synthetic topologies and on the five open-source systems alike — is simulator-derived. The evaluation can establish whether a learned model recovers a simulator’s ordering on architectures it was not trained on. Whether that ordering corresponds to which components actually fail in production is a question this design cannot answer, and Section 8.3 treats it as the study’s principal construct-validity threat.

5. The Explanation Layer: Standards-Grounded Criticality Attribution

The predictor described in Section 4 identifies the concentration of risk but does not address remediation strategies. A component may be deemed critical due to its status as an unreplicated single point of failure,

its role as an error-propagating cascade hub, or as a high-coupling maintainability bottleneck. Each of these structural causes necessitates distinct remediation approaches, such as broker replication, circuit breaker insertion, or refactoring of module dependencies. This section formalizes the diagnostic layer responsible for attributing these causes. SaG decomposes component and relationship criticality into a standards-grounded quality profile, calculated using the same typed node properties (Section 3.4) but without parameter sharing with the neural predictor, and applies this profile to flagged components through triage rather than data flow (Figure 1).

The evidential status of this layer is explicitly stated: it represents an unvalidated design contribution intended for qualitative attribution rather than quantitative ranking. As demonstrated in Section 7.1, its standalone rank correlation is low ($\rho = 0.205$, consistently below unweighted centrality), its elicited AHP weights underperform compared to a uniform prior (Section 7.3), and no human-subject studies have yet assessed developer adoption. This layer maps topological properties to standardized ISO/IEC concepts.

5.1. Grounding in ISO/IEC Standards

In accordance with **ISO/IEC 25010:2023** [13] and **ISO/IEC 25019:2023** [14], SaG formalizes two primary criticality dimensions: **Component Criticality** (D_1) (service loss upon component failure) and **Relationship Criticality** (D_2) (service decline upon channel severance).

Criticality is assessed across two orthogonal characteristics: **Reliability** (R) and **Maintainability** (M). Reliability is further divided into **Fault Tolerance** (FT) (error-propagation reachability on $G^\top$, evaluated using Reverse PageRank, in-degree, and cascade depth potential to inform redundancy and circuit breaker strategies) and **Availability** (A) (exposure to single points of failure, evaluated using directed articulation points, bridge ratios, and connectivity degradation to inform replication strategies). Maintainability (M) assesses structural coupling and code-level complexity, utilizing metrics such as betweenness, QoS-weighted fan-out, code quality penalties, and clustering to guide decoupling and refactoring. Supplementary Table S3 provides a summary of this partition, mapping each ISO/IEC sub-characteristic to its corresponding graph metrics and remediation roles. Safety and security considerations that require specialized hazard logs are excluded from purely structural topology analysis.

5.2. Composite Quality Score Formulation

Rank-normalize all raw metrics to $[0, 1]$ within the graph. Quality sub-characteristics are formulated hierarchically using the Analytic Hierarchy Process (AHP) [59]:

1. **Fault Tolerance** ($FT(v)$): Evaluates error cascade potential on transpose graph $G_{\text{analysis}}^\top$:

$$FT(v) = 0.45 \cdot \text{RPR}(v) + 0.30 \cdot \text{Deg}_{\text{in}}(v) + 0.25 \cdot \text{CDPot}_{\text{enh}}(v),$$

where $\text{RPR}(v)$ is Reverse PageRank (RPR), $\text{Deg}_{\text{in}}(v) = d_{\text{in}}(v)/(|V| - 1)$ is normalized in-degree, and $\text{CDPot}_{\text{enh}}(v) = \text{depth}(v)/\max_{u \in V} \text{depth}(u)$ is the normalized cascade depth potential, measuring the longest reachable directed failure-propagation chain from v on $G_{\text{analysis}}^\top$.

2. **Availability** ($A(v)$): Identifies structural single points of failure across five terms:

$$A(v) = 0.2563 \cdot \text{AP}_c^{\text{dir}}(v) + 0.1998 \cdot \text{QSPOF}(v) + 0.1998 \cdot \text{BR}(v) + 0.2563 \cdot \text{CDI}(v) + 0.0878 \cdot w(v) \quad (10)$$

where $\text{AP}_c^{\text{dir}}(v)$ is Directed Articulation Point (AP) severity, $\text{QSPOF}(v)$ is QoS-weighted Single Point of Failure (QSPOF) severity, $\text{BR}(v)$ is Bridge Ratio (BR), $\text{CDI}(v)$ is Connectivity Degradation Index (CDI), and $w(v)$ is intrinsic QoS weight.

3. **Reliability** ($R(v)$): Blends Fault Tolerance and Availability: $R(v) = r_{\text{FT}} \cdot FT(v) + (1 - r_{\text{FT}}) \cdot A(v)$, with $r_{\text{FT}} = 0.36$. The intra-dimension weights apply $\lambda = 0.70$ shrinkage blending with a uniform prior. Because comparison matrices are rank-one by construction (Supplementary Section S4), these weights represent documented conventions rather than independently elicited consensus.
4. **Maintainability** ($M(v)$): Blends structural coupling with static code analysis:
 $M(v) = 0.35 \cdot \text{BT}(v) + 0.30 \cdot w_{\text{out}}(v) + 0.15 \cdot \text{CQP}(v) + 0.12 \cdot \text{CouplingRisk}_{\text{enh}}(v) + 0.08 \cdot (1 - \text{CC}(v))$,
 where $\text{BT}(v)$ is Betweenness Centrality (BT), $w_{\text{out}}(v)$ is QoS-weighted efferent coupling, $\text{CQP}(v)$ is Code Quality Penalty (CQP), and $\text{CC}(v)$ is local Clustering Coefficient (CC).

The baseline composite quality score integrates both dimensions: $Q(v) = 0.80 \cdot R(v) + 0.20 \cdot M(v)$. When evaluated under an ISO/IEC 25019 Context of Use vector $\vec{\omega} = [q_R, q_M]^\top$, the score is dynamically reweighted: $Q_{\text{domain}}(v) = q_R \cdot R(v) + q_M \cdot M_{\text{static}}(v)$. Components are partitioned into Tukey tiers: CRITICAL ($Q > Q_3 + 1.5 \cdot \text{IQR}$), HIGH, MEDIUM, and MINIMAL. A high A combined with low FT indicates a single point of failure that necessitates replication, while high FT identifies an error-cascade hub that requires circuit breakers (Section 8.4).

5.3. Prescriptive Remediation and Counterfactual Verification

After root causes are attributed, automated refactoring operators generate candidate repair manifests, such as broker replication, circuit breaker insertion, or topic decoupling. An `EditVerifier` constructs the mutated graph G' in memory and counterfactually re-simulates multi-threshold cascades. Candidate repairs are accepted only if they reduce systemic impact beyond simulation seed noise ($\Delta \bar{I}_{\text{comp}} > \kappa \cdot \sigma_{\text{seed}}$, $\kappa \geq 1.0$) and do not introduce new articulation points. This counterfactual verification loop is presented to illustrate the architectural pattern linking diagnosis to remediation. The present study does not evaluate standalone empirical claims regarding prescriptive repair efficacy or production patch synthesis, and automated refactoring benchmarks are reserved for future work.

6. Experimental Setup

6.1. Datasets and System Corpus

The evaluation corpus consists of 2,812 components distributed across seventeen system architectures. This includes twelve synthetic topologies, which constitute the inductive cross-validation folds, and five real-world reference systems that are withheld from all training procedures, as detailed in Table 3.

Table 3: Overview of the evaluation corpus. The twelve synthetic topologies correspond to the inductive Leave-One-Scenario-Out folds described in Table 5. The five real-world systems are excluded from all training folds and are used exclusively for zero-shot transfer (Section 7.4). Per-scenario entity and edge counts, obtained from the committed topology files and verified through continuous integration, are provided in Supplementary Section S13.

Regime	$ V $	$ V_{\text{app}} $	Topics	Brokers	Hosts	Libs	$ E $
Synthetic evaluation scenarios (11)	2,387	1,295	588	60	194	250	10,657
Synthetic case study — ATM (1)	74	26	27	5	8	8	261
Synthetic subtotal (12 LOSO folds)	2,461	1,321	615	65	202	258	10,918
Real-world subtotal (5 systems)	351	141	120	16	32	42	700
Total	2,812	1,462	735	81	234	300	11,618

6.1.1. Reproducibility of the Corpus

The benchmark corpus is designed to be fully regenerable rather than merely statically archived. Each dataset is deterministically generated from its configuration file via:

```
python cli/generate_graph.py batch -input-dir data/scenarios -output-dir <path>
```

A companion manifest records the random seed, entity counts, git commit hash, and a SHA-256 cryptographic digest for each emitted topology. Continuous integration regression tests assert that every committed dataset regenerates byte-identically from its configuration and that all disk digests match the manifest. This guarantees that third parties can reproduce the exact graphs used in our experiments, rather than simply sampling from similar distributions.

6.2. Baselines and Evaluated Predictors

We evaluate four primary predictor configurations drawn from three families. Predictor names indicate the family and substrate: an -N suffix denotes a model trained on the native multigraph, its absence denotes the obtained Application—Library flow projection, and a -QoS suffix denotes a configuration that consumes declared QoS contracts. SaG throughout denotes the framework, never an individual predictor.

1. **Heterogeneous graph learning (typed HGT). HGT-QoS** (proposed): relation-specific Heterogeneous Graph Transformer (Section 4) ingesting the complete native multigraph with 16-dimensional continuous-categorical edge features that encode middleware QoS contracts. We report its ablation, HGT, which masks those QoS dimensions, in Section 7.3.1.
2. **Homogeneous graph learning (untyped GAT). GAT-N-QoS**: homogeneous Graph Attention Network [65] trained on the identical native multigraph substrate with per-type input projections, but untyped, single-relation message passing. Its edge channel carries the scalar QoS aggregate $w(e)$ — dimension 0 of the same 16-D encoding HGT-QoS consumes — rather than the per-dimension decomposition; no homogeneous architecture in our suite ingests the full 16-D vector. The HGT-QoS-GAT-N-QoS contrast therefore bounds the joint contribution of relational typing and per-dimension QoS encoding. Section 7.3.1 separates the second factor within the typed architecture, and the corresponding unweighted ablation is GAT-N. The -N suffix denotes the native substrate and is load-bearing: the same homogeneous architecture run on the DEPENDS_ON projection is reported as GAT / GAT-QoS, and Table 4 reports that pair.
3. **QoS-weighted structural baseline (training-free). Topo-QoS**: QoS-weighted topological centrality evaluated on the obtained application flow projection.
4. **Unweighted structural baseline (training-free). Topo**: structural centrality combining unweighted betweenness centrality and articulation point scoring on the flow projection.

In addition, the out-of-distribution evaluation (Table 5) reports RM ($Q(v)$, the deterministic hierarchical quality attribution model of Section 5) as a diagnostic reference baseline. RM is not fitted to rank failure impact; its inclusion demonstrates how much learned relational prediction adds versus static structural attribution (Section 1.2). Furthermore, deterministic RM scoring drives every sensitivity sweep in Section 7.3, where closed-form formulations isolate parameter effects from neural training stochasticity.

6.2.1. Evaluation Substrates and Parity Guarantee

Predictors in this study operate on matched substrates within their respective families:

Graph Learning Models (GAT-N-QoS, HGT-QoS): Under Leave-One-Scenario-Out (Table 5), both learned predictors ingest the complete native typed multigraph across all five entity types (recorded using the shared -N suffix). In-distribution (Table 4), GAT/GAT-QoS consume the Application—Library DEPENDS_ON projection, confounding typing with multi-entity visibility. Node type reaches GAT-N-QoS only through its per-type projection, with untyped GATConv across edges, whereas HGT-QoS uses relation-specific HGTCov weights. The edge channel also differs: GAT-N-QoS consumes a scalar $w(e)$, while HGT-QoS consumes all 16 dimensions (separated in Section 7.3.1). Parameter budget (434,620 vs. 28,168) as well as directionality remain open confounds (Section 8.4).

Training-Free Structural Baselines (Topo, Topo-QoS): We evaluate topological baselines on the obtained Application—Library DEPENDS_ON projection (Section 3.2), since raw multigraphs route messages via topics/brokers, leaving Application nodes with near-zero betweenness.

Ground-Truth Independence Guarantee: Crucially, **no predictor in either group accesses $G_{\text{structural}}$** : that raw topology is strictly reserved for simulation oracles (Section 4.4), verified by `tests/test_independence_guarantee.py`. Regardless of substrate, all variants are scored on an identical, independently resolved Application node set (V_{app} , Section 6.3).

6.3. Evaluation Metrics and Protocols

Figure accessibility. All figures use the Okabe–Ito palette with distinct markers and hatchings for monochrome legibility.

Ranking Precision: Evaluated via Spearman ρ and Kendall τ against the simulated impact $I^*(v)$ from the primary oracle (Section 4.3). **Critical-Set Identification:** Measured via $F_1@K$, $\text{Precision}@K$, and $\text{Recall}@K$ for top- K components ($K = \text{round}(0.20 \cdot |V_{\text{app}}|)$), which coincide identically as top- K set overlap. **Statistical Significance:** Paired Wilcoxon signed-rank tests [80] ($p < 0.05$) and bootstrap 95% CIs ($B = 2,000$) over folds [81, 82]. In the 12-fold LOSO design, the power floor is $p = 0.00049$. Applying Holm’s step-down correction across ten full-population rank contrasts (Sections 7.1–7.3.1), three survive: **Topo-QoS** over **Topo** ($p = 0.0010$), **Topo** over **RM** ($p = 0.0005$), and unweighted typing **HGT** over **GAT-N** ($p = 0.0010$); **QoS-weighted typing** ($p = 0.0122$) and **QoS edge ablation** ($p = 0.0093$) remain nominally significant with 11/12 directional fold consistency. **Pre-registration.** We pre-registered the primary out-of-distribution contrast (**HGT-QoS** vs. **Topo-QoS** under LOSO, 5 fixed seeds, fold as unit of analysis) in the replication package before obtaining results. As reported in Section 7.1, the margin did not reach statistical significance.

6.3.1. Evaluation Population and Protocols

Every predictor within an evaluation table is scored on an identical node population, resolved strictly from scenario topology and ground truth — the **Application** set (V_{app}) unless noted. Pooling node types conflates distinct base rates and triggers Simpson’s paradox (Section 7.3).

In-Distribution Evaluation: Stratified 60% train / 20% val / 20% test node splits over five seeds {42, 123, 456, 789, 2024}, redrawing partitions and initializations. **Inductive LOSO Cross-Validation:** Models are trained on eleven scenarios and test zero-shot on the held-out twelfth across all 12 folds under equal 3-layer depth and inner-split early stopping (Section 8.4). **Real-World Architectural Transfer:** Synthetic-trained models are evaluated zero-shot on five open-source systems without fine-tuning.

7. Results and Empirical Analysis

This section presents empirical results for RQ1–RQ5 across the twelve-fold inductive benchmark and five authentic open-source distributed systems. Evaluated populations are strictly stratified on the Application service set (V_{app}) under the input-label independence guarantee (Section 4.4).

7.1. RQ1: Graph Learning vs. Structural Baselines

Table 4 presents in-distribution held-out Spearman rank correlation (ρ) versus simulated cascade impact $I^*(v)$ across all twelve distributed architecture scenarios ($n = 12$).

7.1.1. Out-of-Distribution (LOSO) Generalization

In inductive Leave-One-Scenario-Out (LOSO) cross-validation, models are evaluated on their capacity to predict cascading criticality over completely unseen system topologies:

Twelve LOSO folds were evaluated, encompassing eleven synthetic scenarios and the ATM case study as detailed in Table 3 (Section 6.1). In each fold, one scenario was held out for zero-shot testing, with the model trained exclusively on the remaining eleven. All model variants were assessed on the same Application node set per fold (Section 6.3), and paired Wilcoxon tests were conducted across the twelve folds, yielding a minimum attainable two-sided p -value of 0.00049. The evaluated populations per fold ranged from 26 to 300 Application nodes, resulting in $K = \text{round}(0.20, |V_{\text{app}}|)$ values between 5 and 60. For $F_1@K$, **HGT-QoS** achieved 0.424 compared to **Topo-QoS**’s 0.388 ($\Delta = +0.037$, prevailing in 7 of 12 folds, $W = 29.0$, $p = 0.470$). The untyped **GAT-N-QoS** attained 0.431 ($\Delta = -0.006$, prevailing in 5 of 12 folds, 1 tie, $W = 27.5$, $p = 0.653$), indicating that critical-set identification does not statistically distinguish the typed model from either untyped learning or the QoS-weighted baseline.

Label-noise ceiling. These correlations are bounded by the reproducibility of the target they are scored against. Re-running the ground-truth oracle across the five seeds gives a test-retest rank correlation between 0.811 and 1.000 across the twelve folds (median 0.982; nine of twelve at or above 0.95), with Microservices the least reproducible at 0.811. **HGT-QoS**’s $\rho = 0.638$ therefore recovers roughly 65% of the attainable signal against the median ceiling, and no predictor in Table 5 can exceed the reproducibility of its own labels.

Table 4: In-distribution held-out Spearman ρ versus simulated cascade impact $I^*(v)$: mean over five seeds with bootstrap 95% CI in brackets; n = held-out Application count. Substrates differ, and the comparison is confounded in-distribution: HGT/HGT-QoS consume the native typed multigraph. In contrast, GAT/GAT-QoS and the topological baselines consume the Application-Library DEPENDS_ON flow projection (Section 6.2). The typed-untyped contrast below therefore mixes typing with multi-entity visibility here; the substrate-matched comparison is the LOSO one in Table 5, where both architectures read the same graph. Each seed redraws the 60/20/20 split and the initialization. The paired significance tests over these scenarios are in Supplementary Section S10.

Scenario	n	Topo	Topo-QoS	GAT	GAT-QoS	HGT	HGT-QoS
ATM System	5	0.538	0.557	-0.393	-0.080	0.492	0.348
AV System	16	0.187	0.797	0.816	0.465	0.637	0.558
Enterprise	60	0.443	0.793	0.779	0.481	0.861	0.878
Financial Trading	12	0.387	0.512	0.565	0.666	0.693	0.730
Healthcare	10	0.291	0.399	0.725	0.575	0.575	0.607
Hub-and-Spoke	14	0.179	0.429	0.363	-0.156	0.421	0.476
Industrial SCADA	28	0.601	0.710	0.656	0.478	0.787	0.839
IoT Smart City	40	0.320	0.397	0.580	0.538	0.849	0.850
Logistics Fleet	22	0.511	0.652	0.746	0.780	0.796	0.815
Microservices	18	0.219	0.344	0.351	0.363	0.141	0.664
Real-Time Gaming	15	0.360	0.802	0.464	0.471	0.651	0.641
Telecom RAN	24	0.402	0.422	0.608	0.350	0.591	0.526
Mean	—	0.370	0.568	0.522	0.411	0.624	0.661

Table 5: Inductive LOSO evaluation, Application population, twelve folds. Learned variants share training set, substrate, depth, and selection rule (Section 6.3), differing in typing and edge channel, and also – as published – regarding parameter budget and message-passing directionality, which Table 7 controls for. Fold score = mean over five seeds; **Fold** σ = spread of the twelve fold means; **Seed** σ = median within-fold spread (zero by construction for deterministic baselines); CI = bootstrap over folds ($B = 2,000$).

Predictor / Reference	Mean LOSO ρ	95% CI	Fold σ	Seed σ	Critical-Set $F_1@K$	Requires Training
<i>Training-free structural baselines</i>						
Topo	0.349	[0.254, 0.452]	0.173	—	0.366	No
Topo-QoS	0.553	[0.443, 0.657]	0.192	—	0.388	No
<i>Learned predictors (shared native substrate, matched training set)</i>						
GAT-N	0.317	[0.254, 0.381]	0.111	0.298	0.328	Yes
GAT-N-QoS	0.604	[0.538, 0.665]	0.112	0.024	0.431	Yes
HGT	0.551	[0.474, 0.617]	0.124	0.114	0.427	Yes
HGT-QoS	0.638	[0.561, 0.710]	0.133	0.052	0.424	Yes
<i>Diagnostic reference — not a ranking model</i>						
RM / $Q(v)$	0.205	[0.092, 0.320]	0.195	—	0.322	No

Top- K critical sets are a much noisier construct: their cross-seed Jaccard has a median of 0.847 and falls to 0.370 (Logistics Fleet), 0.500 (Industrial SCADA), and 0.500 (Telecom RAN). That instability is the main reason the $F_1@K$ margins are less stable than the ranking margins, and it bounds how much weight any single critical-set comparison can carry. Notably, Microservices is both the least reproducible fold and one of the two on which typed learning loses (Section 7.2.1) — part of that deficit may be label noise rather than model failure.

Figure 2 summarizes these results alongside critical-set identification and inter-oracle agreement.

Key Insights concerning RQ1:

1. **Typed learning is the best configuration, but not demonstrably better than the QoS baseline.** HGT-QoS leads all predictors out-of-distribution ($\rho = 0.638$). Against training-free Topo-QoS it is +0.085 (9/12, $W = 20.0$, $p = 0.151$, CI $[-0.029, +0.194]$), an interval that includes zero; un-augmented HGT is indistinguishable from the baseline outright (-0.002 , 3/12, $p = 0.470$). Neither pre-registered contrast reaches significance, and we report that as the answer rather than as a near

miss.

2. **A QoS-weighted structural score is a genuinely strong baseline — and untyped learning is worse than it.** Topo-QoS reaches $\rho = 0.553$ zero-shot, beating unweighted Topo on all twelve folds ($+0.204$, $p = 0.0005$). More pointedly, the untyped, unweighted learned model loses to it decisively (GAT-N, -0.236 , $2/12$, $p = 0.0024$): on this task, a homogeneous graph network trained on eleven architectures does not reach what a closed-form centrality score achieves with no training at all. Any claim that graph learning is required must be made against this baseline.
3. **Critical-set identification does not favor the typed model.** On $F_1@K$, HGT-QoS scores 0.424 against GAT-N-QoS's 0.431 and HGT's 0.427 — a three-way tie within noise — while all three numerically lead Topo-QoS (0.388), though without statistical significance throughout folds ($\Delta = +0.037$, $p = 0.470$). The margin over untyped learning claimed in earlier versions does not hold on the reconciled 12-fold corpus.
4. **Power is not the limiting factor.** At $n = 12$, the design tolerates four lost folds and still reaches $\alpha = 0.05$, provided the losses are smallest in magnitude. HGT-QoS's are not: it loses Enterprise (-0.335) and Telecom RAN (-0.169) to Topo-QoS by the two largest margins in the set, which is what holds W at 20.0. Enlarging the corpus will not resolve this; we must understand the inversions instead (Section 7.2.1).
5. **The explanation layer is weakly predictive, not noise.** RM/ $Q(v)$ reaches $\rho = 0.205$, losing to unweighted Topo on every fold (-0.144), so no ranking claim is made for it. Its interval $[0.092, 0.320]$ stays above zero, and it supplies interpretable diagnostics without training (Section 5). Table 5 lists it as a reference point, not a competitor.

Software-as-a-Graph (SaG) Pre-Deployment Reliability Evaluation — application population

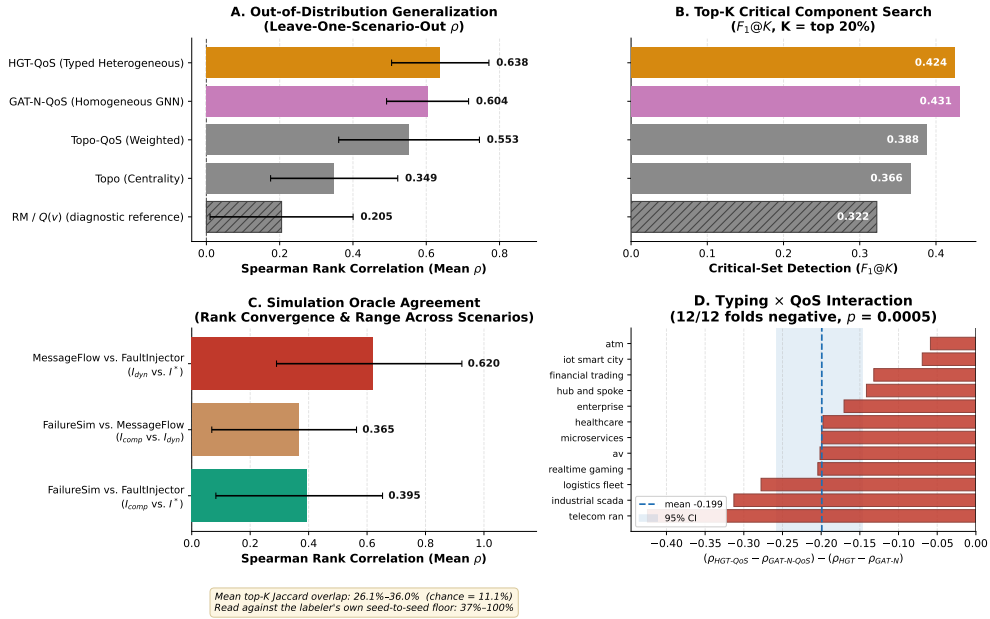


Figure 2: Results at a glance: Application population. (A) Out-of-distribution rank correlation per predictor across the twelve LOSO folds. (B) Critical-set identification at $K = 20\%$, where the typed model does not separate from untyped learning. (C) Pairwise rank agreement between the three simulation oracles, against the chance baseline. (D) The typing $\times$ QoS interaction per fold — how much relation typing buys when the QoS edge channel is present, minus how much it buys when it is absent. Every one of the twelve folds is negative, as the substitution claim of Section 7.2 predicts and the evidence it rests on; the dashed line is the mean and the band is its bootstrap 95% CI. Panels A and B are read from the same artifact as Table 5, C from the convergent-validity artifact, and D from the significance artifact behind Table 7.

7.1.2. The Active Stratum: Ranking Quality Separated From Inertness Detection

Every LOSO figure above shows correlation over the full held-out Application population, and between 21% (Microservices) and 52% (Healthcare) of that population carries exactly zero simulated impact, depending on the fold. A predictor can therefore score well by separating components that can propagate a failure from those that cannot, without ordering the propagating ones correctly. Because these are different capabilities with different functional value, we re-score all twelve folds on the *active stratum* — the $n_{>0}$ components with strictly positive ground-truth impact — using the same predictions, folds, and seeds. Table 6 reports both.

Table 6: LOSO means over twelve folds on the full Application population (ρ) and restricted to components with strictly positive ground-truth impact ($\rho_{>0}$). Same predictions, folds, and seeds; only the evaluated subset differs. Retained is defined as the active-stratum ratio $\rho_{>0}/\rho$ (percentage of full-population correlation preserved). Across all predictors, this restriction retains roughly half the correlation.

Predictor	ρ (full)	$\rho_{>0}$ (active)	Retained
RM / $Q(v)$	0.205	0.102	49%
Topo	0.349	0.181	52%
Topo-QoS	0.553	0.280	51%
GAT-N	0.317	0.159	50%
GAT-N-QoS	0.604	0.328	54%
HGT	0.551	0.299	54%
HGT-QoS	0.638	0.356	56%

Two consequences follow, and the first reverses a claim made in earlier versions of this paper.

- The restriction costs every predictor roughly half its correlation, learned or not.** Retained fractions run from 49% (RM) to 56% (HGT-QoS) with no systematic separation between the training-free and learned families (Topo-QoS 51%, GAT-N 50%, HGT 54%). Roughly half of every predictor’s full-population correlation reflects separating inert components from active ones, a property of the label distribution rather than a discriminator between methods.
- The method ordering is unchanged, and so are the verdicts.** On the active stratum HGT-QoS still leads ($\rho_{>0} = 0.356$), ahead of Topo-QoS (0.280) and GAT-N-QoS (0.328). Nothing in Section 7.1 or Section 7.2 turns on whether zero-impact components are included; we report both columns because they answer alternative operational questions.

7.2. RQ2: Value of Typed Heterogeneity

RQ2 asks whether relation typing boosts prediction over homogeneous message passing. Answering it requires ablating typing against a matched comparator, and the answer depends entirely on which comparator is chosen:

- Both factors have a real main effect.** Averaged over the other factor’s levels, relation typing is worth $\Delta\rho = +0.134$ (12 of 12 folds, $W = 0.0$, $p = 0.0005$, Holm 0.0015) and the QoS edge channel $+0.187$ (11 of 12, $p = 0.0015$, Holm 0.0015). We qualify this: because the HGT and GAT architectures differ in parameter count ($15.4\times$) and message directionality alongside typing, this $+0.134$ margin reflects the joint architectural transition to HGT rather than isolated relational typing alone.
- But they interact, strongly and sub-additively.** The difference of differences — how much typing buys with the QoS channel present, minus how much it buys without it — is -0.199 , negative on all twelve folds ($W = 0.0$, $p = 0.0005$, Holm 0.0015, 95% CI $[-0.258, -0.147]$).
- The simple effects show the same pattern on both sides.** Typing is worth $+0.234$ when the QoS channel is absent (12/12, $p = 0.0005$) and $+0.035$ when it is present ($p = 0.1294$); the QoS channel is worth $+0.287$ without typing ($p = 0.0010$) and $+0.087$ with it ($p = 0.2036$).
- In-distribution fitting (Table 4) is not evidence either way.** The homogeneous pair reads the Application–Library projection while the typed pair reads the native multigraph, confounding message passing with multi-entity visibility.

Typing and QoS encoding are substitutes, not complements. Each mechanism, alone, lifts the plain baseline from $\rho = 0.317$ to roughly 0.55–0.60; together they reach 0.638, barely more than either achieves by itself. Both supply the model with the same underlying information: which relation a message crosses. Relation-specific parameters encode it in the weight matrices; the QoS edge vector encodes it in the edge features. Furthermore, $I^*(v)$ ’s ordering is recovered at mean $\rho = 0.965$ by a topology-only relabeling dropping the QoS term entirely (Section 4.3), confirming that neither channel tracks QoS-driven impact the oracle does not itself express.

What the reference arm is, and why it matters for the effect sizes. Both large simple effects are measured against GAT-N, a floor rather than a competitor ($\rho = 0.317$, losing to Topo-QoS by -0.236 , $p = 0.0024$). Its score carries high seed instability ($\sigma = 0.298$ vs. mean 0.317). The interaction is robust across folds, but read simple effects as recoveries from a deficit rather than absolute gains.

Critical Confounders in the Typing Comparison. While substrate, training set, depth, and selection rules are held constant, two architectural factors remain unmatched: (1) **Parameter Capacity:** HGT-QoS carries 434,620 parameters on the primary training graph vs. 28,168 for GAT-N-QoS ($15.4\times$); and (2) **Message Directionality:** HGT-QoS executes bidirectional message passing (103,725 parameters) enabling downstream nodes to aggregate upstream representations, whereas GAT-N-QoS propagates signals strictly forward. Because $I^*(v)$ measures downstream cascade starvation, bidirectional visibility confers an built-in topological advantage. The observed $+0.134$ typing main effect is thus consistent with relational inductive bias, but equally consistent with capacity or backward edge advantages (Section 8.4). On cost grounds, untyped QoS-weighted GNNs reach $\rho = 0.604$ at 28,168 parameters vs. HGT-QoS’s 0.638 at 434,620 ($15.4\times$ capacity gap for a non-significant margin).

Table 7: The 2×2 over relation typing (T) and the QoS edge channel (Q), whose four cells are the four reported learned arms: GAT-N ($\neg T\text{-}Q$), HGT (T-Q), GAT-N-QoS ($\neg TQ$), HGT-QoS (TQ). **Holm correction is applied across the three orthogonal quantities in the upper block only.** The four simple effects below are algebraically linked to those three — given the cell means, any three determine the fourth — so correcting across them would treat one structural fact as four questions; they are reported descriptively because they carry the narrative, and the claim that they differ from one another rests on the interaction row above, not on the difference between their p -values. Main effects average over the other factor’s levels. **Won** counts folds with $\Delta > 0$; the interaction is *negative* on all twelve, which is the direction the substitution claim predicts. All quantities are post-hoc, and none were pre-registered.

Quantity	Contrast	$\Delta\rho$	Won	W	p	p_{Holm}
<i>The 2×2: three orthogonal quantities, Holm-corrected across these three</i>						
Typing (main effect)	averaged over Q	+0.134	12/12	0.0	0.0005	0.0015
QoS channel (main effect)	averaged over T	+0.187	11/12	2.0	0.0015	0.0015
Typing $\times$ QoS interaction	difference of differences	−0.199	0/12	0.0	0.0005	0.0015
<i>Simple effects — descriptive, not separately corrected</i>						
Typing, QoS absent	HGT vs. GAT-N	+0.234	12/12	0.0	0.0005	—
Typing, QoS present	HGT-QoS vs. GAT-N-QoS	+0.035	9/12	19.0	0.1294	—
QoS channel, typing absent	GAT-N-QoS vs. GAT-N	+0.287	11/12	1.0	0.0010	—
QoS channel, typing present	HGT-QoS vs. HGT	+0.087	10/12	22.0	0.2036	—

7.2.1. Where Typed Learning Fails, and How It Can Be Detected

HGT-QoS loses to Topo-QoS on three folds, but only two of them are substantive: Enterprise ($\rho = 0.569$ vs. 0.838) and Microservices (0.483 vs. 0.563). The third, Real-Time Gaming (0.776 vs. 0.780), is a tie to within 0.004 and carries no interpretive weight. The two substantive inversions that make the RQ1 comparison below significant are both folds on which the training-free baseline is unusually strong, suggesting the model is discarding structural signal the baseline retains rather than failing to learn.

Absence of a Label-Free Confidence Signature. We evaluated whether the standard deviation of predicted scores $\hat{\sigma}$ over held-out applications may indicate model trustworthiness at inference time without labels. Although the two worst-performing folds (Enterprise, $\hat{\sigma} = 0.111$; Microservices, 0.138) exhibit low dispersion, the correlation does not hold across the full benchmark: $\hat{\sigma}$ correlates with the margin over Topo-QoS at

$\rho_s = -0.126$ ($p = 0.697$) for HGT-QoS, and the second-lowest dispersion fold (Healthcare, 0.123) yields one of the largest positive margins (+0.284). Neither graph size (rank correlation with margin $-0.357, p = 0.255$) nor connection density reliably flags fold difficulty in advance.

Feature scale drift across scenarios (documented in `results/feature_shift_diagnostic.md`) remains the primary explanation for the Enterprise deficit, as Enterprise is the largest graph (520 nodes) and feature-scaling disparities are most acute there. Because protocol strictly holds model hyperparameters constant across folds, practitioners currently have no automated, label-free signal to predetermine whether an unseen architecture will favor the learned model or the training-free baseline (Section 8.4).

7.3. RQ3: Ablations and Sensitivity Analysis

This section presents ablations relevant to the primary claims, including the QoS edge encoding, cross-oracle agreement, and per-type stratification that informs the interpretation of all subsequent results. Parameter-sensitivity sweeps over the explanation layer’s ten declared weight constants are provided in the supplementary material (Supplementary Sections S1–S2) to demonstrate robustness rather than to establish new findings. Of the ten constants, only the Fault-Tolerance/Availability blend r_α and the AHP shrinkage λ exhibit appreciable influence on ρ ($\mu^* = 0.144$ and 0.117 under Morris screening, compared to ≤ 0.023 for the remaining eight). No configuration of the topic-weight or QoS sub-weight constants alters any comparison reported above. The elicited AHP weights are, notably, *anti*-predictive: rank correlation decreases monotonically from 0.319 under a uniform setting to 0.200 under raw AHP judgment. These weights are retained because RM serves as an attribution instrument rather than a ranking model, a trade-off discussed in Section 8.4.

7.3.1. QoS Feature Ablation

To isolate the specific empirical contribution of the continuous-categorical QoS edge features (Section 4.1.1), we evaluated **HGT**, an unaugmented ablation of HGT-QoS whose edge features contain only scalar coupling and relation one-hot encodings.

Under the inductive LOSO evaluation, the QoS edge encoding’s value depends on whether relation typing is already present. This is not a second finding but the same one seen from the other side: the interaction tested in Section 7.2 (-0.199 , negative on all twelve folds, $p = 0.0005$) is symmetric in the two factors. Hence, a conditional effect of typing on the QoS channel is necessarily also a conditional effect of the QoS channel on typing. The figures below are the simple effects of Table 7 restated for the ablation reader; the significance of their *difference* rests on that interaction, not on the difference between their p -values.

Without typing, the encoding is decisive. GAT-N-QoS reaches $\rho = 0.604$ against GAT-N’s 0.317: $\Delta\rho = +0.287$, won in 11 of 12 folds, $W = 1.0$, $p = 0.0010$, Holm-corrected $p = 0.0029$, CI $[+0.207, +0.365]$. *With typing, it is not significant.* HGT-QoS reaches 0.638 against HGT’s 0.551: $+0.087$, 10 of 12 folds, $W = 22.0$, $p = 0.2036$, Holm-corrected $p = 0.2588$. An earlier version of this manuscript reported the typed gain as $+0.054$ at $p = 0.0093$ from a superseded artifact and treated it as an independent contribution on top of typing; the effect does not reproduce at that significance, and the independence claim is withdrawn.

The encodings also improve optimization reproducibility, and here the asymmetry runs the other way. The median within-fold standard deviation across five seeds is 0.024 for GAT-N-QoS against 0.298 for GAT-N — more than a tenfold reduction — and 0.052 for HGT-QoS against 0.114 for HGT. An untyped model without the QoS channel is the least stable configuration by a wide margin, and either mechanism stabilizes it. This is consistent with the ranking result: both channels tell the model which relation an edge belongs to, and a model given neither must infer it from topology alone.

The target is nearly QoS-free, which bounds what either channel can credit.. The gains above are earned against $I^*(v)$, whose ordering a topology-only relabeling recovers at mean $\rho = 0.965$ across the same twelve folds, with no QoS term in the labeler at all (Section 4.3). The QoS edge channel therefore cannot help the model track QoS-driven impact that the oracle does not itself express, which is the strongest evidence we have for reading it as a relation-identity channel rather than a contract-semantics one. The label does move under QoS, but only at its top- K boundary (Jaccard 0.678 against the topology-only arm), not in its ranking. A corpus whose oracle expressed QoS-driven impact in its ordering — deadline misses, durability replay,

priority inversion under load, none of which I^* observes — would be a stronger test of the encodings than the one we report.

QoS Parameter Variance. Modal QoS shares range from 29% to 89% across the twelve scenarios (Supplementary Section S5), ensuring that every fold carries genuine variation in declared reliability, durability, and priority. As noted in Section 4.1.1, one schema dimension (`max_blocking_ms_log`) remains zero throughout the corpus as a reserved extension point, while the declared deadline populates the other two (`has_deadline`, `deadline_ns_log`) on 75% of topics; reported gains therefore stem from six active dimensions.

7.3.2. Convergent Validity Over Simulation Oracles

The three reliability-facing oracles measure distinct constructs, so we checked whether they agree before treating any as ground truth. Over the twelve inductive folds on the Application population, the behavioural queue-flow oracle and the topological cascade injector agree at mean Spearman $\rho = 0.620$ (top- K Jaccard 0.365 against 0.111 expected by chance), against I^* ’s own seed-to-seed test-retest of 0.811–1.000. The agreement is therefore substantial but distinctly below label noise, which is the reading we want: an oracle reproducing another to within its own reproducibility would be re-measuring the same topology rather than corroborating it. Two boundaries qualify this — a large share of the agreement is the two oracles concurring on which components are *harmless*, and I_{dyn} has a measured noise floor of its own that the headline does not correct for. Supplementary Section S9 reports the full pairwise table, the zero-excluded correlations, and the multi-seed floors for both non-primary oracles.

7.3.3. Node-Type Stratification and Attention

One result governs how every other number in this paper is read. Measured against $I_{\text{comp}}(v)$ over the eight scenarios of the detection benchmark, stratified RM rank correlations are $\rho = 0.566$ (Application), 0.119 (Broker) and 0.244 (Node), while pooling all types collapses the correlation to $\rho = 0.098$ — below every per-type value it aggregates, which is Simpson’s paradox in its textbook form. This is why we report every evaluation on a single stratum, and why pooled critical-set figures should be read as inflated wherever they appear. Supplementary Section S6 reports the rule-based anti-pattern catalog evaluated on the same benchmark; it summarizes that the catalog flags 93.8% of scored components and therefore does not discriminate, so critical-set identification is delegated to the continuous rankers of Sections 7.1–7.2.

Aggregated by relation type over the ATM case study, first-layer mean HGT attention orders **USES** into libraries (0.227) above publish–subscribe channels (0.163–0.176), but the spread across all eight relation types is narrow (0.15–0.23) and driven substantially by destination in-degree artifacts. Supplementary Section S8 gives the layer-wise distribution and heatmap, and confirms that typed attention remains active across relation types without establishing a statistically distinct ordering.

7.4. RQ4: Real-World Distributed Architecture Validation

We evaluated the framework on five open-source distributed systems transcribed from public repositories: Online Boutique, Train-Ticket, Home Assistant, Autoware Universe (ROS 2), and EdgeX Foundry. All carry labels from the same simulation oracles used throughout, testing topological transfer rather than agreement with field failures.

We conduct two distinct real-world evaluations. First, we evaluate the closed-form explanation layer $Q(v)$ against $I_{\text{comp}}(v)$ in Supplementary Section S7, achieving strong correlation across all five systems ($\rho = 0.514$ –0.800) and outperforming degree centrality. Both training-free references are scored here. **Topo-QoS** was absent from earlier versions of Table 8 because the open-source adapters carried no QoS contracts; this was a defect in how we assessed its betweenness rather than a property of the data, and Section 8.4 records the correction.

7.4.1. Zero-Shot Transfer of the Learned Model

To test generalization to architectures outside our generator, we trained **HGT-QoS** on all twelve synthetic scenarios. We evaluated it zero-shot across the five open-source systems versus $I^*(v)$ (five seeds). To

reduce cross-scenario feature-scale drift, this transfer evaluation applies within-graph rank normalization, 2 message-passing layers, and 150 epochs. No real-world system contributed training gradients or was used to 982 select checkpoints. Table 8 presents the resulting transfer performance.

Table 8: Zero-shot transfer to five open-source systems, scored against $I^*(v)$ on the Application population. HGT-QoS trains on all twelve synthetic scenarios ($\pm$ = spread over five seeds); RM and Topo are training-free and are scored on identical labels and nodes. $\rho_{>0}$ restricts the correlation to the $n_{>0}$ components that actually propagate a failure and is the column to read for ranking quality; full-population ρ conflates that with separating active from inert. Both training-free references are scored on the same labels and nodes (Section 8.4).

Real-World Architecture	$ V_{app} $	RM ρ	Topo ρ	Topo-QoS ρ	HGT-QoS ρ	HGT-QoS $\rho_{>0}$	$n_{>0}$	$F_1@K$
Cloud Microservices Mesh	22	0.777	0.891	0.888	0.649 ± 0.131	-0.029	18	0.400
Train-Ticket Booking Mesh	41	0.713	0.528	0.541	0.776 ± 0.004	-0.213	22	0.450
Autoware.universe (ROS 2)	32	0.357	0.307	0.378	0.734 ± 0.054	+0.559	28	0.633
EdgeX Foundry (Industrial IoT)	22	0.470	0.534	0.534	0.804 ± 0.040	+0.304	19	0.500
Home Assistant (Smart Home)	24	0.265	0.297	0.289	0.872 ± 0.035	+0.704	23	0.600
Mean	—	0.516	0.511	0.526	0.767	+0.265	—	0.517

Key Insights for Real-World Transfer:

- The full-population figure is not a ranking result.** On all Applications, HGT-QoS reaches $\rho = 0.767$ vs. 0.511 (Topo), 0.526 (Topo-QoS), and 0.516 (RM), leading on 4/5 systems. However, between 4% (Home Assistant) and 46% (Train-Ticket) of Applications carry zero simulated impact; full correlation heavily rewards separating inert from active components rather than ranking active ones.
- Restricted to components that actually propagate failures, transfer is not established.** On the active stratum the mean drops to +0.265, and both microservice call trees invert: Cloud Microservices ($\rho_{>0} = -0.029$, $n = 18$) and Train-Ticket (-0.213 , $n = 22$). The three pub-sub systems hold up (+0.559 Autoware, +0.704 Home Assistant, +0.304 EdgeX). **We therefore report RQ4 as a negative result:** learned relational transfer to authentic open-source architectures is not established.
- The QoS-weighted baseline is now scored, and it sharpens one comparison.** Topo-QoS reaches $\rho = 0.888$ on Cloud Microservices, where the learned model scores worst (0.649) and inverts on active components, sharpening the deficit on synchronous call trees.
- What the full-population number does support.** Separating propagating components from non-propagating ones narrows the review surface effectively, reflected in the $F_1@K$ column (0.517 mean, 0.633 on Autoware).
- Architectural Ingestion Boundary.** The active-stratum inversion on microservices emphasizes a fundamental domain mismatch: synchronous RPC architectures propagate chain failures backward along invocation trees via timeout accumulation and thread pool starvation [45], whereas asynchronous pub-sub architectures cascade forward via queue saturation and topic starvation. Because HGT-QoS was trained only on pub-sub communication semantics, its relational inductive bias inverts when applied to synchronous call trees. We therefore recommend a strict ingestion boundary: practitioners should deploy SaG’s learned GNN pipeline on asynchronous and event-driven architectures (ROS 2, Kafka, DDS, MQTT), and rely on closed-form structural baselines (Topo-QoS, $\rho = 0.888$) or dedicated static call-graph analyzers for synchronous RPC/REST microservice meshes.

7.5. RQ5: Analysis Cost and Its Comparison Against Simulation

RQ5 quantifies computing overhead and sustainability during CI/CD evaluation, with per-stage latencies reported in Table 9:

The neural model constitutes the least expensive stage, whereas the deterministic model incurs substantial computational cost. For 2,000 components, the HGT forward pass requires 56 ms, compared to 239 s for deterministic structural analysis, yielding a ratio of $4,259\times$. This ratio reflects cost within the pipeline rather than the total cost of architectural evaluation. Indices 0–17 of each node feature vector (Section 3.4)—including betweenness, closeness, reverse PageRank, articulation, and bridge scores—are generated during this analysis stage, making the forward pass dependent on its completion. End-to-end evaluation of an unseen 2,000-component architecture requires approximately four minutes, with the learned model accounting for only

Table 9: Per-stage latency of the inference pipeline across scaling graph sizes (CPU, median of 3 runs).

$ V $	$ E $	Analyze (s)	Graph→tensor (s)	HGT forward (ms)	Analyze : forward
249	1,127	1.74	0.010	26.5	66×
499	2,402	8.32	0.022	16.4	509×
999	6,422	44.54	0.056	21.1	2,108×
1,998	19,301	239.34	0.157	56.2	4,259×

0.02% of this time; the 56 ms figure represents the marginal cost of re-scoring an already-analyzed graph. Across the eight-scenario detection benchmark, the complete gate (structural analysis plus 18 anti-pattern detectors) executes in 0.04–82.7s, with the upper bound corresponding to the 520-component Enterprise mesh.

One metric dominates cost, and it grew. Measured cost now tracks the stage’s $O(|V|^2 + |V||E|)$ bound closely: from 249 to 1,998 components wall-clock rises 138× against a 137× growth in $|V||E|$. The dominant term is the Connectivity Degradation Index, which is computed for every node in the main connected component rather than for articulation points alone. That choice is deliberate and is a correctness requirement rather than an oversight: gating CDI to articulation points leaves it identically zero for every node whose removal does not literally disconnect the graph, which drives $A(v)$ to a near-constant in the redundant multi-publisher topologies this system targets. The cost is the price of a non-degenerate Availability score, and we report it rather than the cheaper gated variant we could have measured.

7.5.1. The Gate Is Not Cheaper Than the Simulation It Replaces

The framing that motivated this analysis — static gating as a low-cost substitute for dynamic simulation — does not survive measurement against our own oracle. Timing the **FaultInjector** labeling sweep (five seeds, node types Application/Broker/Library, the full ground-truth run) on the same corpus and the same idle hardware gives 0.14–7.2s per scenario, against 0.04–82.7s for the analysis gate. Both maxima occur in the 520-component Enterprise mesh, so the largest scenario compares 7.2s of simulation against 82.7s of static analysis: the gate costs roughly eleven times more than the simulation it is meant to replace.

This finding refutes the assumption that static analysis is computationally cheaper than in-process simulation: breadth-first cascade traversal is simpler than computing all-pairs connectivity degradation ($O(|V|^2 + |V||E|)$). However, in practical continuous integration workflows, static SSA provides a key deployment trade-off: (1) it scores components (e.g., shared libraries, hosts) and dependency edges that node-level simulation passes do not evaluate; and (2) deterministic graph metrics can be incrementally cached across git commits, recomputing only the k -hop neighborhood touched by an architectural pull request. We clarify that the benchmark timings in Table 9 reflect full from-scratch recomputation without caching; once cached, GNN scoring executes in 56 ms, whereas repeating full simulation sweeps requires re-running stochastic traversals globally. Without such caching, direct simulation is strictly faster.

8. Discussion, Threats to Validity, and Limitations

8.1. Discussion and Practical Consequences

When to use Topo-QoS, and when to use HGT-QoS. The results do not justify a straightforward recommendation of the learned model over the closed-form alternative; therefore, the trade-offs are presented based on empirical measurements rather than expectations.

- 1. Training-free ranking (Topo-QoS).** The training-free ranking approach (Topo-QoS) obviates the need to train the model, checkpoint storage, or retraining, and achieves $\rho = 0.553$ in zero-shot testing across twelve synthetic architectures and 0.526 across five open-source systems. Learned models do not provide a statistically significant improvement in ranking performance (+0.085, $p = 0.151$, with the confidence interval including zero; and 0.888 versus 0.649 in favor of the baseline on Cloud Microservices). Consequently, Topo-QoS works as a robust default for scalar criticality ordering.

2. **One learned mechanism, not two (GAT-N-QoS or HGT).** Implementing either relation typing or a QoS edge channel individually produces considerable improvements over an untyped, unweighted model (+0.234 and +0.287, respectively; Holm-corrected $p \leq 0.0015$), whereas combining both mechanisms yields minimal additional benefit. Untyped QoS-weighted graph neural networks (GNNs) offer a more favorable efficiency trade-off, achieving $\rho = 0.604$ with 28,168 parameters compared to HGT-QoS's 0.638 with 434,620 parameters. In contrast, unweighted homogeneous models (GAT-N, $\rho = 0.317$) perform worse than training-free heuristics, indicating that GNNs lacking relational signals are not effective for this task.
3. **Capabilities without a closed-form counterpart.** Certain capabilities are unattainable using closed-form methods. Typed relational attention enables identification of the specific channels mediating cascades (Supplementary Section S8), and edge-level criticality (I_{edge} , Eq. (9)) facilitates assessment of individual dependencies for circuit-breaker placement, hence extending analysis beyond node-level rankings. These features provide substantive design justifications for adopting typed models when additional computing overhead is acceptable.

Dual-Engine Consensus Protocol. A previous version of this study proposed an automated tiered fallback based on prediction dispersion $\hat{\sigma}$. However, this heuristic does not replicate: $\hat{\sigma}$ correlates negatively with the margin over Topo-QoS at $\rho_s = -0.126$ (Section 7.2.1), leading to the withdrawal of that recommendation. Instead, since both engines execute within seconds, SaG now provides a dual-mode (**-predictor-mode dual**) scoring manifest with HGT-QoS and Topo-QoS concurrently. This approach flags unanimous top- K components for immediate remediation and highlights substantial ranking divergences for human architectural review, rather than concealing them behind an uncalibrated threshold.

Role of the Explanation Layer. The RM attribution profile ($Q(v)$, Section 5) offers transparent, standards-compliant architectural diagnostics in accordance with ISO/IEC 25010. By distinguishing single-point-of-failure exposure (Availability) from broad fault propagation reach (Fault Tolerance), RM delivers qualitative remediation guidance, such as determining whether a component requires replication or decoupling, which purely numeric rankers and simulation oracles are unable to provide.

8.2. Performance and Computational Sustainability Implications

What sustainability means for a pre-deployment gate. Green software engineering considers the energy consumption associated with development and assurance, in addition to execution [29, 83, 84, 85, 86, 30, 31]. Chaos engineering and staging fault injections require significant cluster-hours per sweep and necessitate provisioned virtual machines, container runtimes, and network emulators. Pre-deployment static manifest analysis eliminates the considerable carbon and financial costs associated with provisioning physical cloud staging clusters and live chaos harnesses. Nevertheless, determining precise energy reductions necessitates empirical measurement using hardware counters [30].

The efficiency claim we withdraw, and where the cost actually sits. The previously stated efficiency claim is withdrawn. The pipeline incurs computational costs in both local developer environments and in-process continuous integration (CI) runners. Static analysis does not reduce raw CPU computation compared to simulation; for example, global connectivity degradation (82.7s on Enterprise) is approximately eleven times slower than breadth-first search (BFS) cascade traversal (7.2s). No general claim is made regarding in-process computational effectiveness relative to simulation. The primary computational expense arises from calculating CDI across all connected nodes, rather than only articulation points, to prevent degenerate Availability scores. In continuous integration, functional computational sustainability is achieved through deterministic graph caching: by caching base graph metrics across git commits and re-extracting features only for pull-request delta subgraphs, gating latency is reduced to the sub-second neural forward pass (56 ms). Consequently, static gate sustainability relies on graph-algorithm optimization and caching, rather than on machine-learning overhead.

8.3. Threats to Validity

Construct Validity. Ground-truth impact $I^*(v)$ derives from discrete-event cascade simulation on structural models rather than live outages. While I^* exhibits substantial correlation with dynamic queue flow I_{dyn} ($\rho = 0.620$ against a 0.811–1.000 label test–retest ceiling, Supplementary Section S9), top- K Jaccard reaches only 0.27–0.37 due to non-linear thresholding. Furthermore, $I^*(v)$ is recovered at $\rho = 0.965$ by topology-only relabeling, reflecting topological reachability rather than dynamic buffer drops. We measure no oracle against production incident telemetry, forming our primary construct boundary.

Internal Validity. Feature leakage is prevented by strict graph separation: predictors consume G_{analysis} , while simulation oracles traverse $G_{\text{structural}}$ (CI-asserted). We maintain parity via matched training sets, depths, and early stopping. Capacity (434,620 vs. 28,168) along with directionality remain open confounds (Section 8.4). In the QoS schema, six active dimensions govern profiles, with one reserved extension point.

External Validity. Evaluation covers twelve synthetic scenarios and five open-source systems. Zero-shot transfer does not hold on active components (mean $\rho_{>0} = +0.265$, inverting on microservice call trees, Section 7.4.1). As Zhou et al. [45] document, microservices cascade backward along call trees via RPC timeouts and retry storms, whereas pub-sub cascades forward via message starvation. GNN directional inductive biases must be conditioned on communication synchrony. Scaling to $> 2,000$ nodes requires incremental graph caching or mini-batching (GraphSAINT [87]).

Conclusion Validity. Heavy-tailed distributions are evaluated using non-parametric correlations (Spearman ρ , Kendall τ), bootstrap CIs ($B = 2,000$), and Wilcoxon signed-rank tests. All analyses are strictly stratified to prevent Simpson’s paradox (pooled $\rho = 0.098$ vs. per-type 0.119–0.566). Zero-excluded metrics isolate ranking from inertness detection. Folds share ten training graphs, and synthetic graphs derive from a single generator family, bounding empirical generalizability.

8.4. Limitations and Future Work

Correction of the Real-World Baseline. Earlier versions omitted **Topo-QoS** from Table 8, explaining this to missing QoS contracts in open-source adapters. However, all five adapters declare QoS parameters. The omission resulted from computing betweenness on the raw multigraph rather than on the **DEPENDS_ON** projection. After this correction, **Topo-QoS** is now reported across all systems.

Explanation Layer Validation. SaG separates Availability from Fault Tolerance, but human studies have not validated practitioner actionability. Furthermore, elicited AHP weights perform worse than a uniform prior at ranking (Section 7.3); we next prioritize counterfactual mutation tests and user evaluations.

Uncontrolled Confounds in Typing. Table 7 holds substrate, training set, depth, and early stopping constant, but parameter budget (434,620 vs. 28,168) and reverse message-passing directionality (103,725 parameters in **HGTConv**) remain unmatched. Because $I^*(v)$ is a downstream-reachability functional, upstream visibility confers an advantage unrelated to typing. To thoroughly separate these factors, we have registered three specific control variants in the SaG benchmark suite (`saag/evaluation/variant_registry.py`): **GAT-N-C** (capacity-matched homogeneous baseline expanded to $\approx 434\text{k}$ parameters), **GAT-N-QoS-C** (capacity-matched homogeneous with QoS edge encoding), and **HGT-QoS-U** (unidirectional HGT with forward message passing only). Executing these registered control arms across all twelve LOSO folds represents the primary empirical priority for subsequent benchmark iterations.

Model Selection and Caching. Early stopping uses an inner validation split on the primary graph; held-out scenario validation is a prioritized extension. Prediction dispersion does not reliably signal OOD fallback (Section 7.2.1). Production deployment requires incremental graph caching over PR diffs to amortize $O(|V|^2 + |V||E|)$ feature extraction.

Future Directions: Distributed AI, Power Testbeds, and Self-Healing. Key extensions include: (1) modeling distributed LLM serving backbones (vLLM, DeepSpeed); (2) measuring hardware energy directly via RAPL/NVML to benchmark static gating against live chaos sweeps in joules; and (3) advancing from predictive diagnostics to prescriptive synthesis, generating automated pull requests with circuit breakers and broker replicas.

9. Conclusion

This study introduced **Software-as-a-Graph (SaG)**, a pre-deployment static analysis framework for asynchronous and event-driven distributed systems. SaG integrates a relation-specific Heterogeneous Graph Transformer for failure-impact forecasting with an interpretable ISO/IEC 25010 Reliability and Maintainability attribution layer. Both components operate on a typed multigraph derived from Architecture-as-Code manifests and require no runtime telemetry.

The central empirical finding is that the framework’s two architectural mechanisms function as substitutes rather than complements, the core claim of this study and observable only through factorial ablation. Relation typing and the 16-dimensional QoS edge encoding each demonstrate a main effect under inductive distribution shift ($\Delta\rho = +0.134$ and $+0.187$, Holm-corrected $p = 0.0015$). However, their interaction is -0.199 and negative across all twelve folds ($p = 0.0005$): typing contributes $+0.234$ to a model without the QoS channel and $+0.035$ to one that includes it. Either mechanism alone recovers most of the full model’s performance; adding the second yields minimal additional benefit. This suggests that both mechanisms encode the same underlying information, specifically which relation a message traverses. The oracle supports this interpretation, as a topology-only relabeling recovers $I^*(v)$ ’s ordering at mean $\rho = 0.965$ without any QoS term. Consequently, neither channel can track QoS-driven impact not already expressed in the ground truth.

This carries a direct practical consequence: a team requiring only scalar criticality rankings should adopt the simpler untyped QoS-weighted baseline ($\rho = 0.604$ at 28,168 parameters vs. HGT-QoS’s 0.638 at 434,620, a non-significant margin for a $15.4\times$ capacity gap). We propose the typed architecture because relation-specific attention and edge criticality (I_{edge} , Eq. (9)) provide structural diagnostic functions that untyped models lack.

The boundary conditions are also clearly delineated: learned ranking does not significantly outperform unparameterized QoS-weighted centrality ($+0.085$, $p = 0.151$), and zero-shot transfer decreases to $+0.265$ on active components, with inversion observed on microservice call trees. Prediction dispersion does not replicate as an out-of-distribution fallback indicator, and elicited AHP weights yield poorer ranking than a uniform prior.

This work contributes a reproducible typed-multigraph formulation of pub-sub architecture, a corpus that regenerates byte-identically from committed configurations, and an evaluation of where two widely assumed architectural inductive biases are beneficial and where they fail to compose. Additionally, the proposed pre-deployment pipeline features a learned component that is three orders of magnitude less costly than its deterministic counterpart, which is itself eleven times more expensive than the simulation it was designed to replace. The core question remains whether such a pipeline predicts failures that actually occur, rather than those produced by a simulator. Addressing this issue will require field data that a static corpus cannot provide.

Declarations

CRedit Authorship Contribution Statement. **Ibrahim Onuralp Yigit:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Feza Buzluca:** Conceptualization, Methodology, Validation, Writing – review and editing, Supervision.

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Supplementary Material. Parameter sensitivity analyses (OFAT, Morris screening, threshold and normalization sweeps, zero-inflation diagnostics, and HGT attention distributions) are provided in the online supplementary document (Sections S1–S8).

Data Availability. The complete replication package—datasets, configurations, simulation harnesses, adapters, checkpoints, and scripts—is available on Zenodo under DOI 10.5281/zenodo.14922108 [88], complete with virtual environment specifications (`uv` and `pip/requirements.txt`) for zero-friction reproduction. Synthetic datasets regenerate byte-identically. All 295 reported table values are deterministically verified against backing JSON artifacts via `reproduce/reconcile_manuscript.py`, which asserts clean git trees for declared artifacts across table cells and named prose quantities.

Declaration of Generative AI and AI-Assisted Technologies. During preparation, the authors used Anthropic’s Claude to assist with LaTeX typesetting and readability. The authors reviewed and edited all content and take full responsibility. No generative AI was used to design the study, analyze research data, or generate experimental results; all figures and tables are rendered deterministically.

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